

DP-600

Fabric Analytics

Engineer Associate

The July 2026 blueprint in full, the Fabric concepts behind every bullet explained properly, and 72 model questions with answers and reasoning — including the Direct Lake, OneLake security and lifecycle detail this exam actually turns on.

100 min

proctored, with interactive components

700 / 1000

scaled passing score, no negative marking

45–50%

of the exam is a single domain:
Prepare data

**SQL • KQL •
DAX**

all three are named in the audience profile

00 Start here

What DP-600 is really testing, why it defeats strong Power BI developers, and how to spend the time.

DP-600 is **not the Fabric version of PL-300**, and treating it that way is the single most common reason experienced Power BI people fail it. PL-300 is a report-and-model exam. DP-600 is a **data platform exam with a semantic-model tail**: almost half of it — 45–50% — is *Prepare data*, which means choosing between data stores, building a star schema inside a lakehouse or warehouse, and querying with SQL, KQL and DAX. The DAX-heavy part you are already good at is only a quarter of the paper.

The blueprint is the specification, not a summary. Microsoft publishes 41 itemised skill bullets across three domains, and the exam is written directly from them. Section 2 reproduces every bullet with what it is testing. Two bullets in particular are worth memorising because they are unusually specific and almost nobody prepares for them: *"Choose between Direct Lake on OneLake and Direct Lake on SQL analytics endpoint"* and *"Perform impact analysis of downstream dependencies from lakehouses, warehouses, dataflows, and semantic models"*.

0.1 The exam at a glance

Mechanics

Exam code	DP-600
Credential	Microsoft Certified: Fabric Analytics Engineer Associate
Level	Intermediate, role-based
Duration	100 minutes (seat time is longer)
Passing score	700 on a 1–1000 <i>scaled</i> scale — not 70%
Format	Proctored, with interactive components
Languages	English, Japanese, Chinese (Simplified), German, French, Spanish, Portuguese (Brazil)
Price	USD 165 list; India list price in the region of ₹4,865 — verify at booking
Renewal	Free, annual, unproctored and open book on Microsoft Learn; window opens 6 months before expiry

Audience profile, as Microsoft words it

"Subject matter expertise in designing, creating, and managing analytical assets, such as semantic models, warehouses, or lakehouses." Responsibilities: **prepare and enrich data for analysis, secure and maintain analytics assets, implement and manage semantic models.**

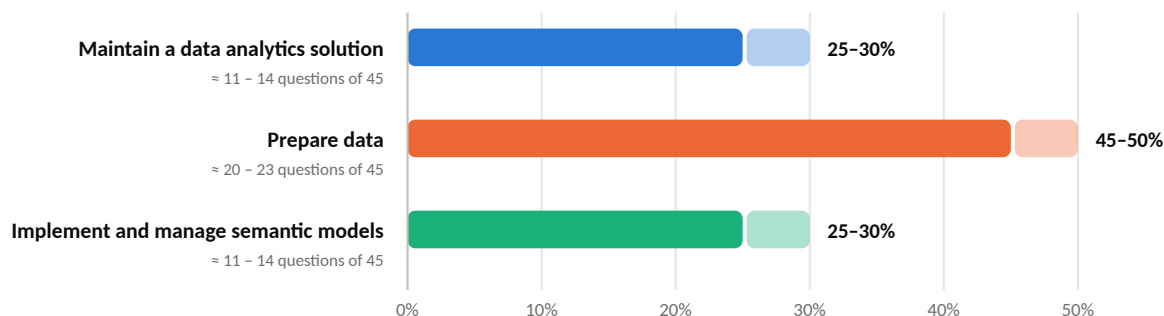
And then the sentence that sets the bar: *"You should also be able to query and analyze data by using Structured Query Language (SQL), Kusto Query Language (KQL), and Data Analysis Expressions (DAX)."*

Read that literally. KQL is not decorative — there is a dedicated blueprint bullet for it, and it is the language most candidates have never written a line of. Two evenings of KQL is the cheapest marginal score improvement available on this exam.

0.2 Where the marks are

Skills measured, by domain

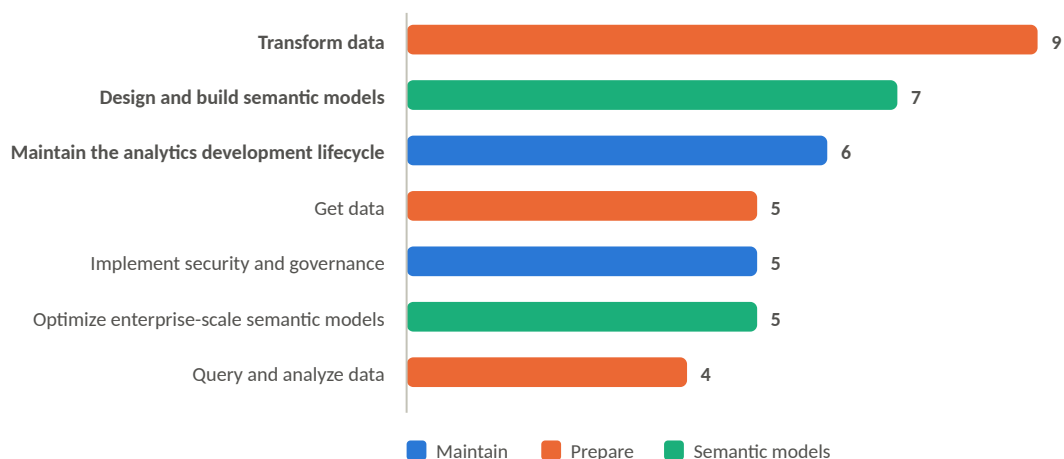
Published weightings as of 21 July 2026; question estimates assume a 45-question form



One domain is roughly the size of the other two combined. If you build a study plan that gives equal time to three domains, you have already mis-allocated about a third of your effort.

Published skill bullets per sub-area

41 bullets in total across the three domains, July 2026 blueprint



The blueprint is compact — 41 bullets against PL-300's 78 — but each bullet is broader. "Choose between different data stores" is one bullet and a week of study.

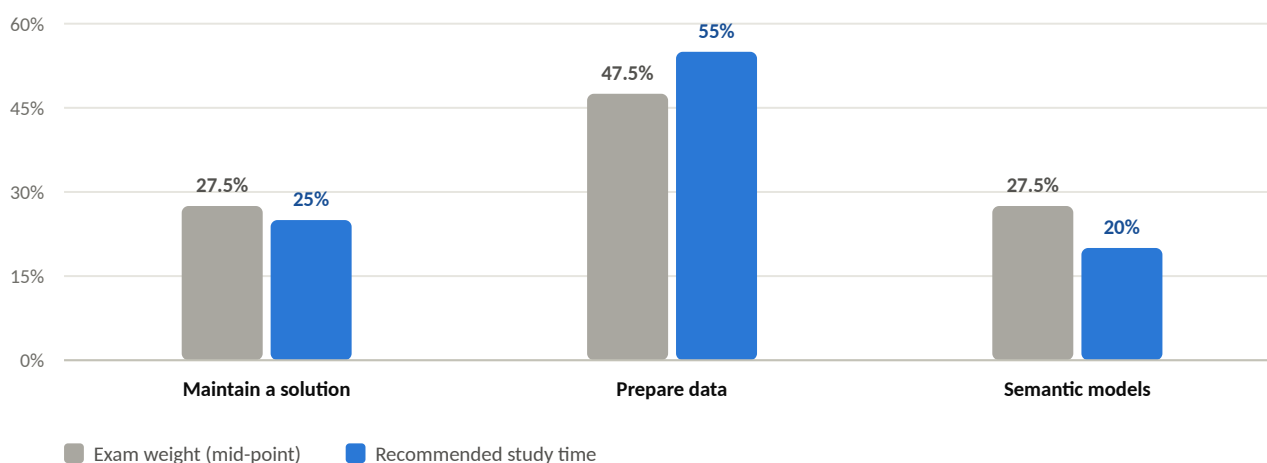
0.3 Why strong Power BI developers fail this exam

TRAP	WHAT IS ACTUALLY BEING TESTED
Treating Fabric as "Power BI with a lakehouse"	The exam expects you to choose between a Lakehouse, a Warehouse, an Eventhouse and a SQL database on the basis of skill set, data type, transaction requirements and write support — not preference. See §3.2.
Hand-waving Direct Lake	There are two distinct Direct Lake variants with materially different capabilities, and a whole blueprint bullet devoted to choosing between them. Fallback exists in one and not the other. See §3.6.
Never having written KQL	A named blueprint bullet. You need to read and write a basic <code> where</code> / <code> summarize</code> / <code> project</code> pipeline and know <code>external_table()</code> . See §3.5.
Assuming Power BI security answers	Fabric adds OneLake security roles, file-level access, item permissions (Read / ReadData / ReadAll / Write) and domains on top of workspace roles. RLS defined at the SQL analytics endpoint is <i>not</i> the same as RLS in the semantic model. See §3.9.
Skipping the lifecycle bullets	Six bullets — Git for workspaces, .pbip projects, deployment pipelines, impact analysis, XMLA deployment, and reusable .pbit/.pbids/shared models. Together they are close to 15% of the exam and are pure recall.
Optimising the model, not the table	Direct Lake performance is a <i>Delta table</i> problem — V-Order, file compaction, row-group size — before it is a DAX problem. See §3.7.

0.4 A study plan weighted to the exam, not to your comfort zone

Exam weight vs recommended study time

Calibrated for someone who already writes advanced DAX and builds enterprise semantic models



The semantic-model domain is deliberately *under-weighted* relative to the exam: if you already own star schemas, DAX and RLS, you need only to top up on Direct Lake configuration, calculation groups, field parameters, composite models and large-model format. Everything you save goes into Prepare data.

WEEK	FOCUS	DELIVERABLE THAT PROVES YOU ARE READY
1	Fabric foundations and the data-store decision	A trial-capacity workspace with a Lakehouse and a Warehouse side by side. Load the same data into both and write down, in your own words, why you would put gold in one rather than the other.
2	Prepare data, part 1 — ingestion and shortcuts	An internal shortcut, an external shortcut to ADLS/S3, a mirrored database, and a Dataflow Gen2 with a lakehouse destination. Be able to say when each is right and what each costs.
3	Prepare data, part 2 — transformation and querying	A gold star schema built with T-SQL views and stored procedures in a Warehouse and with PySpark in a Lakehouse. Then answer the same business question four ways: Visual Query Editor, SQL, KQL, DAX.
4	Semantic models on Fabric	Two Direct Lake models over the same gold layer — one on OneLake, one on the SQL analytics endpoint. Deliberately trip a guardrail on each and observe the different failure modes.
5	Optimisation	A before/after on a badly-fragmented Delta table: file count, average file size, row groups, and the query timing difference. Plus one DAX measure tuned with the DAX query view.
6	Security, governance and ALM	Workspace roles vs item permissions vs OneLake security roles demonstrated on one lakehouse; sensitivity labels; endorsement; a Git-connected workspace; a three-stage deployment pipeline with rules; an impact-analysis screenshot.
7	Practice and gaps	Official practice assessment twice, with everything under 75% revisited. Then sit the exam.

Use the official practice assessment and the exam sandbox. Microsoft publishes both free on the certification page. The sandbox shows you the actual interface and question widgets, which matters more on DP-600 than on PL-300 because of the interactive components. Neither is a substitute for hands-on time in a real Fabric capacity — a trial capacity is free and is the single best preparation resource that exists.

On brain dumps. Sites selling "real DP-600 questions" breach the exam NDA and can void your certification permanently. Every question in Section 4 of this guide is original, written to the published blueprint and to documented product behaviour.

01 Question formats, scoring and exam-day tactics

Short section, real marks — particularly the review rules and the way DP-600 stems are constructed.

1.1 What the paper looks like

TYPE	WHAT IT LOOKS LIKE	HOW TO HANDLE IT
Single-answer multiple choice	Four or five options, one correct. The bulk of the exam.	Find the constraint clause before you look at the options. On DP-600 it is usually one of: <i>minimise data movement, minimise administrative effort, without duplicating data, the solution must support T-SQL writes, least privilege</i> .
Multiple response	"Select two" or "select three", scored per component.	Partial credit applies, so answer even when one of the pair is a guess. Never select more than asked.
Drag and drop / build list	Order steps, or match features to requirements. Common for deployment-pipeline setup, shortcut creation, and security layering.	Sequence answers follow the real product workflow. Walk the UI in your head rather than reasoning abstractly.
Hot area / dropdown completion	A code block — SQL, KQL, DAX or PySpark — with dropdowns, or a clickable UI image.	This is where the SQL/KQL/DAX requirement in the audience profile actually bites. You must <i>read</i> all three languages fluently even if you write only one.
Yes/No repeated-scenario series	The same scenario three times with a different proposed solution each time; each scored independently and you cannot return once you move on .	Judge each on its own merits. Two of three can be correct.
Case study	A multi-tab scenario — existing environment, business requirements, technical requirements — followed by several questions.	Read the requirements tabs first. You may review <i>within</i> the case; once you leave it, it is locked.

1.2 Scoring

Scaled, not percentage

1–1000, adjusted for form difficulty. 700 is the bar and it does not correspond to a fixed number of questions. Do not try to compute how many you can afford to lose.

Partial credit, no penalty

Multi-part items score per correct component and there is no negative marking. An unanswered question and a wrong one score identically, so **never leave one blank**.

Unscored items exist

Some questions are being trialled and do not count toward your score. You cannot tell which — treat everything as scored and do not let one strange item unsettle you.

1.3 Tactics that reliably help on this exam specifically

- **"Without duplicating the data"** means a **shortcut**. This is the most reliable keyword mapping on the whole paper. "Near-real-time replica of an operational database with no pipeline" means mirroring.

- **"Must support T-SQL INSERT / UPDATE / DELETE" means Warehouse.** A Lakehouse's SQL analytics endpoint is read-only. This single fact decides several questions.
- **"Must enforce row-level security defined in SQL" means Direct Lake on the SQL analytics endpoint** — Direct Lake on OneLake reads the files directly and bypasses it.
- **Least privilege wins in every security question.** Viewer plus a OneLake security role beats Contributor; ReadData beats ReadAll; an item permission beats a workspace role.
- **Prefer the platform feature over the hand-built one.** Deployment pipelines over manual copying, impact analysis over a spreadsheet of dependencies, mirroring over a custom CDC pipeline.
- **Watch the direction of "upstream".** DP-600 repeatedly rewards fixing a problem at the Delta table rather than in DAX. If a Direct Lake model is slow and one option is **OPTIMIZE**, look hard at it.
- **If two options are identical in effect, both are wrong.** Re-read the stem; you have missed a qualifier.

02 The syllabus, bullet by bullet

All 41 published skills in the 21 July 2026 blueprint, with what each one is actually testing.

Blueprint status. The skills measured are those effective **21 July 2026**; the study-guide page was last revised in June 2026. The only change in that revision was a *minor* update to **Optimize enterprise-scale semantic models** — which is where the Direct Lake on OneLake vs SQL analytics endpoint choice now sits explicitly. The three domains and their weightings did not change. Anything you study that predates the 2025 Direct Lake split is out of date on the highest-value bullets.

1 Maintain a data analytics solution

25–30%

Implement security and governance

- **Implement workspace-level access controls** — Admin / Member / Contributor / Viewer, and which of them can manage OneLake security or write data. See §3.9.
- **Implement item-level access controls** — sharing an individual lakehouse, warehouse or semantic model, and the Read / ReadData / ReadAll / Write permission set.
- **Implement row-level, column-level, object-level, and file-level access control** — four distinct mechanisms in Fabric, at two different layers. This is the single densest bullet in the domain.
- **Apply sensitivity labels to items** — Microsoft Purview labels; they classify and protect and follow the data into exports, but they grant nothing.
- **Endorse items** — Promoted vs Certified vs Master data, who can do each, and what it changes (discoverability, not permissions).

Maintain the analytics development lifecycle

- **Configure version control for a workspace** — Fabric Git integration with Azure DevOps or GitHub; branch per workspace, commit and update flows.
- **Create and manage a Power BI Desktop project (.pbip)** — TMDL for the model, PBIR for the report; why this is what made real code review possible.
- **Create and configure deployment pipelines** — 2 to 10 stages, default 3; item pairing; deployment rules to repoint sources per stage.
- **Perform impact analysis of downstream dependencies from lakehouses, warehouses, dataflows and semantic models** — lineage view and the item's Impact analysis pane; the answer whenever a stem asks "before making a breaking change".
- **Deploy and manage semantic models by using the XMLA endpoint** — Tabular Editor / SSMS / ALM Toolkit, metadata-only deployment, and the read-write XMLA setting on the capacity.
- **Create and update reusable assets, including .pbip template files, .pbids data source files, and shared semantic models** — three different reuse mechanisms; know exactly what each one contains.

2 Prepare data

45–50%

Get data

- Create a data connection — cloud connections, gateways for on-premises, and who may bind a connection to a shortcut.
- Discover data by using the OneLake catalog and Real-Time hub — the discovery surfaces for stored and streaming data respectively.
- Ingest or access data as needed — the four routes: pipelines (Copy activity), Dataflow Gen2, notebooks, and **access without ingestion** via shortcuts or mirroring.
- Choose between different data stores — Lakehouse vs Warehouse vs Eventhouse vs SQL database. See §3.2; expect several questions from this one bullet.
- Implement OneLake integration for Eventhouse and semantic models — making KQL/Eventhouse data available in OneLake as Delta so a Direct Lake model can read it.

Transform data — 9 bullets; the densest sub-area on the exam

- Create views, functions and stored procedures — T-SQL in a Warehouse (full DDL/DML) versus a Lakehouse SQL analytics endpoint (views only, read-only).
- Enrich data by adding new columns or tables · Denormalize data · Aggregate data · Merge or join data
- Implement a star schema for a lakehouse or warehouse — surrogate keys, SCD handling, and the fact that this is a *gold layer* activity, not a semantic-model one.
- Identify and resolve duplicate data, missing data or null values — in SQL (`ROW_NUMBER` de-duplication), in PySpark (`dropDuplicates` , `fillna`), and in Dataflow Gen2.
- Convert column data types · Filter data

Query and analyze data

- Select, filter and aggregate data by using the **Visual Query Editor** — the no-code Power Query-style surface on a lakehouse or warehouse; it generates SQL.
- Select, filter and aggregate data by using **SQL** · by using **KQL** · by using **DAX**
- Four bullets, four languages. Expect at least one code-reading question in each of SQL, KQL and DAX.

3 Implement and manage semantic models

25–30%

Design and build semantic models

- Choose a storage mode — Import, DirectQuery, Direct Lake (two variants), Dual, and composite.
- Implement a star schema for a semantic model · Implement relationships, such as bridge tables and many-to-many relationships
- Write calculations that use DAX variables and functions, such as **iterators**, **table filtering**, **windowing** and **information functions** — "windowing" means `INDEX` , `OFFSET` , `WINDOW` , `RANK` , `ROWNUMBER` ; "information functions" means `ISBLANK` , `HASONEVALUE` , `SELECTEDVALUE` , `ISINSCOPE` , `USERPRINCIPALNAME` .
- Implement calculation groups, dynamic format strings and field parameters
- Identify use cases for and configure **large semantic model storage format**
- Design and build composite models — and know which Direct Lake variant permits them.

Optimize enterprise-scale semantic models

- Implement performance improvements in queries and report visuals — Performance Analyzer, visual count, reducing returned rows.
- Improve DAX performance — DAX query view, storage-engine vs formula-engine split, avoiding context transition in large iterators.
- **Configure Direct Lake, including default fallback and refresh behaviour** — the `DirectLakeBehavior` property and framing/automatic updates.
- **Choose between Direct Lake on OneLake and Direct Lake on SQL analytics endpoint** — the highest-value single bullet on the exam. See §3.6.
- Implement incremental refresh for semantic models

03 Concepts, in the detail the exam expects

Eleven deep-dives. Sections 3.2, 3.6 and 3.9 alone account for a large share of the paper.

3.1 OneLake and the shape of Fabric

OneLake is a single, tenant-wide data lake, provisioned automatically, built on ADLS Gen2, in which every Fabric workload stores its data in the **same open format: Delta-Parquet**. A Lakehouse, a Warehouse and an Eventhouse all land Delta tables in the same lake. That is the architectural claim the whole exam rests on: *one copy, many engines*.

- **Workspace** is the unit of collaboration, capacity assignment and Git connection. **Domain** groups workspaces by business area with delegated administration.
- **Capacity** (F SKUs; legacy P SKUs) is the compute you pay for. Every operation consumes Capacity Units; sustained overuse produces throttling after a smoothing window.
- **OneLake catalog** is the discovery surface for stored data; **Real-Time hub** is the discovery surface for streaming sources and events. The blueprint names both.
- Because storage is common, the choice of item is a choice of *engine and skill set*, not of file format. Say that in your head every time a question offers Lakehouse vs Warehouse.

3.2 Choosing a data store — the decision table

	LAKEHOUSE	WAREHOUSE	EVENTHOUSE / KQL DB
Data type	Structured, semi-structured and unstructured	Structured only	Telemetry, logs, time-series, free-text events
Skill set	Data engineers; Spark, PySpark, Scala	SQL developers and citizen developers	Analysts comfortable with KQL
Primary interface	Notebooks and Spark; plus a read-only SQL analytics endpoint	Full T-SQL — DQL, DML and DDL	KQL querysets, Real-Time Dashboards
T-SQL writes	No . The endpoint cannot INSERT / UPDATE / DELETE ; you write with Spark	Yes , fully ACID	Ingestion-oriented; not a T-SQL target
Multi-table transactions	No	Yes	No
Storage	All Delta-Parquet in OneLake. Eventhouse data can be made available in OneLake as Delta via OneLake integration , which is how a Direct Lake model reads it.		
Choose it when	Ingesting messy or mixed data, medallion architecture, Spark-native team, files as well as tables	Enterprise dimensional warehousing, stored procedures, a SQL-native team, transactional guarantees	Sub-second queries over high-volume append-only event data

The two sentences that answer most of these questions. "The solution must support T-SQL UPDATE statements" → Warehouse. "The team writes PySpark and the source includes JSON and images" → Lakehouse. A **Fabric SQL database** is the fourth option and belongs to operational/OLTP workloads whose data is mirrored into OneLake for analytics; a **Power BI datamart** is the legacy self-service option and is rarely the right answer in a Fabric question.

3.3 Getting data in — four routes, and two of them copy nothing

ROUTE	WHAT IT IS	CHOOSE IT WHEN THE STEM SAYS...
Data pipeline	Orchestration: Copy activity, control flow, parameters, schedules and event triggers. The same engine as Azure Data Factory.	"orchestrate", "on a schedule", "copy from 40 tables", "run the notebook after the copy succeeds".
Dataflow Gen2	Power Query with explicit destinations (Lakehouse, Warehouse, KQL DB, SQL DB).	"analysts must maintain the transformation", "low-code", "reuse existing Power Query logic".
Notebook	PySpark / Spark SQL. Testable, version-controllable, scales.	"large volumes", "complex logic", "merge/upsert for slowly changing dimensions", "the team writes Python".
Shortcut	A pointer to data elsewhere — internal (another lakehouse, warehouse, KQL DB, semantic model) or external (ADLS Gen2, Amazon S3 and S3-compatible, Google Cloud Storage, Dataverse, Iceberg, OneDrive/SharePoint, on-premises via gateway). No copy, no latency, schema syncs automatically.	"without duplicating the data" , "without moving the data", "share a gold table across workspaces", "query existing S3 data in place".
Mirroring	Continuous, managed replication of an operational database (Azure SQL DB, Cosmos DB, Snowflake, PostgreSQL and others) into OneLake as Delta.	"near real-time analytics on an OLTP source", "without building a pipeline", "without loading the source system".

Shortcut detail the exam can reach: in a Lakehouse, shortcuts in the **Tables** folder can only be created at the top level and normally point at Delta data, while shortcuts in the **Files** folder can be at any level and point at any format. Access requires permission in *both* the calling item and the target; internal shortcuts authorise with the calling user's identity. Deleting a shortcut never affects the target; deleting or moving the target breaks the shortcut.

3.4 Transforming: building the gold layer

The blueprint's star-schema bullet says "for a lakehouse **or warehouse**" — meaning the dimensional model is built *before* the semantic model, in the data store. Points the exam tests:

- **Surrogate keys** generated at load, not business keys; integer keys compress and join better.
- **SCD Type 2** implemented with a **MERGE** — in T-SQL in a Warehouse, or in PySpark against a Delta table in a Lakehouse. Facts join on the surrogate key so history resolves correctly.
- **Views, functions and stored procedures** are a Warehouse capability. A Lakehouse SQL analytics endpoint supports views but is read-only, so no stored procedure can write.
- **Denormalise deliberately.** Snowflaked dimensions cost a relationship hop in the semantic model; flatten them in gold.
- **Duplicates and nulls** — the three idioms worth having ready:

```

-- T-SQL: keep the latest row per business key
WITH d AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY CustomerKey
                                ORDER BY UpdatedAt DESC, Id DESC) AS rn
    FROM stg.Customer )
SELECT * FROM d WHERE rn = 1;

# PySpark: de-duplicate and handle nulls in a Lakehouse
df = (spark.read.format("delta").load("Tables/stg_customer")
      .dropDuplicates(["CustomerKey"])
      .fillna({"Segment": "Unknown", "CreditLimit": 0}))
df.write.format("delta").mode("overwrite").saveAsTable("dim_customer")

```

3.5 Four query languages, one business question

The blueprint gives each of these its own bullet. You do not have to author fluently in all four, but you must **read** all four and know which surface each belongs to.

SURFACE	WHERE IT LIVES AND WHAT IT IS FOR
Visual Query Editor	A no-code, Power Query-style canvas on a Lakehouse SQL analytics endpoint or a Warehouse. Drag tables, join, filter, group — it generates SQL, which you can view. The answer whenever a stem says "without writing code".
SQL	Warehouse (full T-SQL) or Lakehouse SQL analytics endpoint (read-only). The default querying language of the exam.
KQL	Eventhouse / KQL database, KQL querysets, Real-Time Dashboards. Pipe-based, read-only, optimised for time-series.
DAX	The semantic model. Also available as a query language in the DAX query view and via <code>EVALUATE</code> .

```
-- SQL: revenue by category for the last 30 days
SELECT p.Category, SUM(f.Revenue) AS Revenue
FROM gold.FactSales f
JOIN gold.DimProduct p ON p.ProductKey = f.ProductKey
WHERE f.OrderDate >= DATEADD(day, -30, CAST(GETDATE() AS date))
GROUP BY p.Category
ORDER BY Revenue DESC;

// KQL: the same shape, pipe by pipe
Sales
| where OrderDate > ago(30d)
| summarize Revenue = sum(Revenue) by Category
| order by Revenue desc

// KQL: reading a shortcut inside a KQL database
external_table('MyShortcut')
| take 100

-- DAX as a query language (DAX query view / XMLA)
EVALUATE
    SUMMARIZECOLUMNS (
        'Product'[Category],
        TREATAS ( { "2026" }, 'Date'[Year] ),
        "Revenue", [Total Revenue]
    )
ORDER BY [Revenue] DESC
```

The KQL vocabulary worth memorising: `where`, `project` (select columns), `extend` (add a computed column), `summarize ... by` (group and aggregate), `bin(Timestamp, 5m)` (time bucketing), `ago(1h)`, `top N by`, `join kind=inner`, `take / limit`, and `render timechart`. That list is enough for every KQL question this exam is likely to ask.

3.6 Direct Lake — the highest-value topic on the exam

Direct Lake loads Delta-Parquet column segments **directly from OneLake into the VertiPaq engine on demand**. It gives Import-class query performance without an Import-style data copy. Two mechanisms explain everything else:

Framing — what "refresh" means here

A Direct Lake refresh copies **metadata only**. It analyses the latest Delta table versions and updates the model's file references. It takes seconds and costs almost nothing in capacity terms. There is no data copy, which is why Direct Lake removes the refresh window entirely. Automatic updates can trigger reframing without a schedule.

Transcoding — what happens at query time

When a DAX query runs, only the columns and row groups it needs are converted from Delta-Parquet into VertiPaq format and cached in memory. Because loading is on demand, **the data analysed can exceed the capacity's maximum memory** — a genuine advantage over Import that the exam likes to state as a fact.

The two variants — and the bullet that asks you to choose

	DIRECT LAKE ON ONELAKE	DIRECT LAKE ON SQL ANALYTICS ENDPOINT
Reads	Delta tables from any Fabric source, directly from the files	A single lakehouse or warehouse, through its SQL endpoint
Composite models	Supported — can mix with Import tables	Not supported
Calculated columns / tables	Supported (calculated columns in preview)	Not supported
SQL views	Cannot connect to them	Supported
SQL-defined RLS	Bypassed — files are read directly	Respected (may fall back to DirectQuery to honour it)
DirectQuery fallback	None . A guardrail breach makes the refresh fail and the model unqueryable until the Delta tables are optimised	Automatic — the query silently runs as DirectQuery instead, slower, with a warning in refresh history
Calculation groups, field parameters, what-if	Supported in both	
Choose it when	IT-controlled analytics over a large central lakehouse; you want composite flexibility and calculated objects; security is handled outside SQL	SQL-defined row- or column-level security must be enforced; you need SQL views; you want automatic resilience

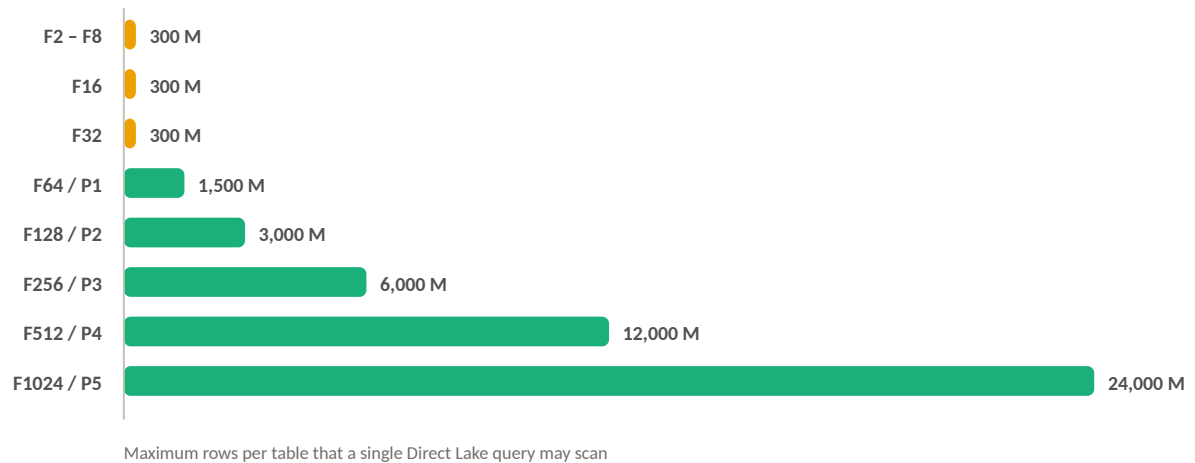
Fallback behaviour, and why you might switch it off

The semantic model property `DirectLakeBehavior` controls what happens when Direct Lake on SQL cannot serve a query. With fallback **enabled** (the default) the query silently reverts to DirectQuery — resilient, but users experience a slowdown nobody notices. With fallback **disabled** the query fails with an error. The documented recommendation is to keep fallback on in production and turn it off in development so regressions are visible. Common fallback triggers: a non-materialised SQL view, SQL-based granular access control, a guardrail breach, or an unprocessed table.

The guardrails

Direct Lake per-query row limit, by capacity SKU

Maximum rows a single query may scan per table; other guardrails cover Parquet files and row groups



F2 through F32 all share the same 300-million-row per-query ceiling; the step change comes at F64/P1. Max model size is evaluated at model level (10 GB at F2–F8, 20 GB at F16, 40 GB at F32, unlimited from F64/P1); every other guardrail is evaluated per query.

Two facts worth carrying into the exam. Direct Lake requires a Fabric capacity — it does not work on a Free licence, in My Workspace, or through any gateway. And a one-side relationship column containing duplicate values causes Direct Lake queries to *fail*, not merely to give odd totals: related columns must have matching data types and the one side must be genuinely unique.

3.7 Delta table maintenance — the other half of "optimise"

Direct Lake performance is a file-layout problem before it is a DAX problem. Excessive small files create more VertiPaq column segments and raise transcoding overhead; the documented sweet spot is **1 to 16 million rows per row group**, with an adaptive target file size in the **128 MB – 1 GB** band.

OPERATION	WHAT IT DOES AND WHEN TO RUN IT
V-Order	A write-time optimisation of Parquet row-group distribution, encoding and compression. Costs roughly 15% more on write, materially faster on read. Note the current default: V-Order is disabled by default in new Fabric workspaces and is enabled per session, per table (<code>delta.parquet.vorder.enabled</code>) or per write. Enable it for read-heavy analytics tables; leave it off for write-heavy staging.
Auto compaction	The recommended default for Spark-written tables: evaluates fragmentation after each write and compacts only when needed. Removes the need for a separate health check.
OPTIMIZE	Bin-compaction — merges small files toward the adaptive target size, and purges a file's deletion vectors once more than 5% of its records are marked deleted. Use it as a one-off on an existing fragmented table, then rely on auto compaction; also run it periodically where updates and deletes accumulate deletion vectors without producing small files.
Z-Order	Co-locates related information in the same files for the columns you filter on most. Complementary to V-Order.
VACUUM	Deletes files no longer referenced by the table, reclaiming storage. It does not improve the layout of active files. Schedule it separately from OPTIMIZE, and do not shorten the retention period without checking time-travel requirements and concurrent readers — removing files early makes required table versions unavailable.

```
# Detect a small-file problem before it becomes a Direct Lake problem
d = spark.sql("DESCRIBE DETAIL gold.fact_sales").first()
print(d["numFiles"], d["sizeInBytes"] / d["numFiles"] / 1024**2, "MB avg")
```

The diagnostic signal to remember: **file count rising faster than active table size, with files staying below the target** means you need **OPTIMIZE**.

3.8 Storage modes, composite models, large models and incremental refresh

MODE	CHOOSE IT WHEN	COST
Import	Data fits comfortably; refresh latency acceptable; you need the full DAX surface and calculated objects.	Refresh windows, memory, a second copy of the data.
DirectQuery	Real-time requirement, or the data must not be copied, or the source is authoritative.	Source latency on every visual; restricted DAX.
Direct Lake	Data already sits as Delta tables in OneLake and you want Import-class speed with no refresh.	Fabric capacity; guardrails; a hard dependency on well-maintained Delta tables.
Dual	A dimension shared by an Import aggregation table and a DirectQuery fact.	Only meaningful inside a composite model.
Composite	Mixing storage modes in one model — including Direct Lake on OneLake plus Import.	Complexity; and it is unavailable on Direct Lake on SQL.

Large semantic model storage format. Without it a semantic model is capped at 1 GB. With it, the model can grow to the capacity limit (or an admin-set maximum), which is required for models beyond 10 GB — the Power BI Desktop *upload* limit stays 10 GB, but the model may grow past it in the service after refresh. Microsoft recommends enabling it for any model you deploy through the **XMLA endpoint**, even a small one, because XMLA write performance improves. It can be set per model, or as the workspace default storage format, and it needs a Fabric F, Premium P, Embedded A or PPU capacity.

Incremental refresh still works the same way: `RangeStart` and `RangeEnd` parameters (exact names, Date/Time type), a filter on the fact query that **folds** to the source, and an archive plus incremental window on the table. Optional: detect data changes to skip unchanged partitions, and a hybrid table with a DirectQuery partition on the leading edge. It applies to Import tables — a Direct Lake table needs no incremental refresh at all, because framing replaces refresh.

3.9 Security and governance — two planes, four data mechanisms

Fabric separates a **control plane** (what you may do) from a **data plane** (what you may see). Getting the two straight answers most security questions on the exam.

Control plane — workspace roles

ROLE	MANAGE ADMINS	MANAGE MEMBERS	MANAGE ONELAKE SECURITY	WRITE DATA	READ DATA	DELETE WORKSPACE
Admin	✓	✓	✓	✓	✓	✓
Member	—	✓	✓	✓	✓	—
Contributor	—	—	—	✓	✓	—
Viewer	—	—	—	—	via role	—

Admin, Member and Contributor already have full data access, so **OneLake security roles only constrain Viewers and users holding an item Read permission**. That is the Fabric restatement of the familiar Power BI fact that RLS does not apply to workspace Members.

Control plane — item permissions

Sharing an individual item grants a permission set: **Read** (metadata), **ReadData** (query through the SQL analytics endpoint, in delegated mode), **ReadAll** (OneLake access via the default reader role), and **Write** (everything). Least privilege is nearly always the intended answer: grant ReadData rather than ReadAll, and an item permission rather than a workspace role.

Data plane — the four granular mechanisms

MECHANISM	WHERE IT IS DEFINED AND WHAT IT RESTRICTS
Row-level (RLS)	Rows. Definable in a OneLake security role, at the SQL analytics endpoint, or in the semantic model (DAX role with <code>USERPRINCIPALNAME()</code>). Which layer is authoritative depends on the path the user takes — this is exactly why the Direct Lake variant matters.
Column-level (CLS)	Columns. In a OneLake security role or at the SQL endpoint. Not authorable in Power BI Desktop.
Object-level (OLS)	Whole tables or measures, hidden entirely from the user. Semantic-model concept; authored with Tabular Editor, not Desktop.
File / folder-level	OneLake security roles scoped to folders, tables or schemas within an item. This is the genuinely new one relative to Power BI.

Sensitivity labels come from Microsoft Purview, apply to reports, dashboards, semantic models, lakehouses and dataflows, and **persist into exports** to Excel, PowerPoint and PDF with their protection intact. They classify and protect; they never grant or remove access. **Endorsement** is separate again: **Promoted** can be applied by anyone with write access, **Certified** only by users the tenant admin authorises, and **Master data** marks the authoritative source for a business entity. Endorsement changes discoverability and trust signalling, not permissions.

3.10 The analytics development lifecycle

CAPABILITY	WHAT YOU MUST BE ABLE TO STATE
Git integration	A workspace connects to an Azure DevOps or GitHub repository and branch. You commit changes from the workspace and update the workspace from the repo. It is version control — not promotion between environments.
.pbip projects	Power BI Desktop saved as a project: the model as TMDL and the report as PBIR , both plain text. This is what makes diffs and pull-request review of a semantic model possible.
Deployment pipelines	2 to 10 stages , 3 by default (Development, Test, Production). Items are paired between adjacent stages and paired items overwrite on deployment; unpaired items with the same name create duplicates instead. Deployment rules repoint data sources and parameters per stage so you promote the same artefact rather than rebuilding it. Deployment is <i>promotion</i> , complementary to Git, not a replacement for it.
Impact analysis	Available on lakehouses, warehouses, dataflows and semantic models: shows the downstream items and reports that depend on the one you are about to change, and lets you notify their owners. The answer whenever a stem says "before making a breaking change" or "identify affected reports".
XMLA endpoint	Read/write access to the semantic model from external tools — Tabular Editor, SSMS, ALM Toolkit, scripted deployment from Azure DevOps. Lets you deploy a metadata-only model change without republishing the report. Requires the capacity's XMLA endpoint to be set to Read Write.
Reusable assets	.pbip — a report template: everything except the data. .pbids — a data source file that opens Desktop pre-pointed at a source. Shared semantic model — one governed model that many reports connect to live, the strongest of the three and the one that pairs with endorsement.

3.11 The DAX the blueprint actually names

The bullet reads: "Write calculations that use DAX variables and functions, such as **iterators**, **table filtering**, **windowing**, and **information functions**." Those four categories are the syllabus.

```
-- Iterators + variables: value per order, not per line
Avg Order Value =
AVERAGEX ( VALUES ( Sales[OrderID] ), CALCULATE ( SUM ( Sales[LineAmount] ) ) )

-- Table filtering: percent of what the user can see
% of Visible =
DIVIDE ( [Revenue], CALCULATE ( [Revenue], ALLSELECTED ( 'Product' ) ) )

-- Windowing: prior row without EARLIER, and a running total
Prev Period = CALCULATE ( [Revenue], OFFSET ( -1, , ORDERBY ( 'Date'[Month], ASC ) ) )
Running Tot = CALCULATE ( [Revenue],
    WINDOW ( 1, ABS, 0, REL,
        SUMMARIZE ( ALLSELECTED ( 'Date' ), 'Date'[Date] ),
        ORDERBY ( 'Date'[Date], ASC ) ) )
Rank = RANK ( DENSE, ALLSELECTED ( 'Product'[Name] ),
    ORDERBY ( [Revenue], DESC ) )

-- Information functions: control totals and identity
Safe Rank = IF ( HASONEVALUE ( 'Product'[Name] ), [Rank] )
Level Label = IF ( ISINSCOPE ( 'Geo'[City] ), "City", "Region" )
Current User = USERPRINCIPALNAME ()
```

The window functions — `INDEX` , `OFFSET` , `WINDOW` , `RANK` , `ROWNUMBER` — share the parameters `relation` , `ORDERBY` , `PARTITIONBY` , `MATCHBY` , `blanks` and (in visual calculations) `reset` . Either `relation` or `ORDERBY` must be supplied.

Calculation groups, dynamic format strings and field parameters are a single blueprint bullet and are regularly confused with each other:

- **Calculation group** — items containing `SELECTEDMEASURE()` that apply to whichever measure is in the visual. Solves the CY/PY/YoY/YTD explosion.
- **Dynamic format string** — a per-item or per-measure DAX expression returning a format string, so "YoY %" renders as a percentage while "PY" renders as currency.
- **Field parameter** — a generated table that lets a user swap which measure or which dimension a visual uses, from a slicer. It changes the field, not the calculation.

04 Model question bank

72 original questions in the exam's formats, with the answer and why every distractor fails.

How to use this bank. Cover the green panel and commit to an answer before you read it — and force yourself to say why each *other* option is wrong. On DP-600 the distractors are usually things that genuinely work, just not under the stem's constraint, so the ability to name the disqualifying clause is the skill being trained. Questions follow the blueprint's proportions: 18 / 30 / 19, plus a five-question case study.

1 Maintain a data analytics solution

Q1 – Q18

Q1

SINGLE ANSWER Analysts must query a lakehouse's gold tables but must not be able to create, edit or delete any item in the workspace, and must only see rows for their own region. What do you configure?

- A. Workspace Contributor role plus a OneLake security role.
- B. Workspace **Viewer** role plus a OneLake security role with a row filter.
- C. Workspace Member role plus RLS on the semantic model.
- D. Workspace Admin role plus a sensitivity label.

ANSWER — B

Viewer is the only workspace role that cannot write. It is also the only role that OneLake security roles actually constrain — Admin, Member and Contributor already have full data access and bypass them.

A and C both grant write access and both bypass the data-plane filter, so the regional restriction would do nothing. D is worse still, and a label grants and removes nothing.

Q2

SINGLE ANSWER An external consultant must query a warehouse through its SQL analytics endpoint but must not be given direct OneLake file access. Which item permission do you grant?

- A. Read
- B. **ReadData**
- C. ReadAll
- D. Write

ANSWER — B

ReadData is precisely "may query through the SQL analytics endpoint". ReadAll is the one that grants OneLake access via the default reader role — the thing the stem forbids.

A Read gives metadata visibility only, so the consultant could see the item but not query it. D grants everything. Least privilege is the deciding principle on almost every security question in this domain.

Q3**YES / NO SERIES**

A OneLake security role with a row filter has been created on a lakehouse. A manager reports that she still sees every row. She holds the workspace Contributor role. Proposed explanation: the role's member list is missing her security group. Is this the most likely cause?

ANSWER — NO

Contributor already has full read and write access to workspace data, so OneLake security roles do not apply to her at all. Adding her to the role would change nothing; the fix is to reduce her workspace role to Viewer, or to have her consume through an item permission instead.

Other proposed explanations in this series and their verdicts: "she is also a workspace Admin" (Yes, same mechanism); "the filter references a column that does not exist" (Yes, plausible); "the lakehouse has no sensitivity label" (No — labels are unrelated to row visibility).

Q4**SINGLE ANSWER**

Within a single lakehouse, one team may read the **Files/hr/** folder and another may not, while both may read all Tables. What do you use?

- A. Two separate lakehouses.
- B. A **OneLake security role scoped to the folder**, granting read to one group only.
- C. Row-level security on the semantic model.
- D. Object-level security in Tabular Editor.

ANSWER — B

OneLake security roles can be scoped to folders, tables and schemas within an item — this is the file/folder-level access control the blueprint names, and it is the mechanism that has no equivalent in classic Power BI.

A solves it by duplication, which the exam consistently penalises. **C** filters rows in a semantic model and has no effect on files. **D** hides tables or measures inside a model, not folders in a lake.

Q5

DRAG AND DROP — MATCH EACH REQUIREMENT TO A MECHANISM

Match the four granular access-control mechanisms to the requirements.

REQUIREMENT	MECHANISM
Sales reps must see only their own accounts' transactions	Row-level security
The Salary column must be invisible to non-HR analysts, but the rows must still be returned	Column-level security
The entire Payroll table and its measures must not appear in the field list for most users	Object-level security
A folder of scanned contracts in the lakehouse Files area must be readable only by Legal	File / folder-level security

ANSWER — AS TABULATED

The distinction the exam relies on: RLS filters *rows*, CLS hides *columns* while still returning rows, OLS removes *tables and measures* from the model entirely so they do not even appear, and OneLake file security governs *files and folders*.

Also remember where each is authored. RLS: OneLake security role, SQL endpoint, or a DAX role in Desktop. CLS: OneLake security role or SQL endpoint. OLS: external tools such as Tabular Editor — **not** Power BI Desktop. File-level: OneLake security roles only.

Q6

MULTI-SELECT — CHOOSE TWO

Which two statements about sensitivity labels in Fabric are correct?

- A. They are defined in Microsoft Purview and applied to Fabric items such as semantic models, lakehouses, reports and dataflows.
- B. The label and its protection persist when data is exported to Excel, PowerPoint or PDF.
- C. Applying a Confidential label removes a user's ability to open the item.
- D. Labels are the supported way to restrict which rows a user can see.
- E. Only a Fabric administrator can apply a label.

ANSWER — A AND B

Persistence into exports is the reason labels exist: classification and encryption follow the data out of Fabric.

C confuses classification with permission — access is governed by workspace roles, item permissions and OneLake security roles.

D is RLS. **E** is false; a user with edit permission on the item and a licence for the label can apply it.

Q7

SINGLE ANSWER

A finance semantic model must be marked as the organisation's authoritative source, and only a nominated review board may award that status. What do you configure?

- A. Promote the semantic model.
- B. **Certify** the semantic model, with the review board configured as the authorised certifiers.
- C. Apply a Highly Confidential sensitivity label.
- D. Move the model to a dedicated domain.

ANSWER — B

Certification is restricted to users the tenant administrator authorises — which is exactly the "only a nominated board" requirement. Promotion, by contrast, can be applied by anyone with write access.

C classifies and protects but says nothing about authority. **D** groups workspaces for administration and discovery; it does not endorse an item. Note the third endorsement level worth knowing: **Master data**, for the authoritative source of a business entity.

Q8

SINGLE ANSWER

A large enterprise wants each business unit to administer its own workspaces, apply its own default sensitivity label and appear as a separate area in the OneLake catalog, without giving anyone tenant-admin rights. What do you implement?

- A. One workspace per business unit with Admin roles assigned.
- B. **Fabric domains**, with workspaces assigned to each domain and domain admins delegated.
- C. Separate Fabric capacities per business unit.
- D. Deployment pipelines per business unit.

ANSWER — B

Domains group workspaces by business area and support delegated administration and domain-level settings — the governance construct that sits above the workspace.

C separates compute and cost, not governance. **A** does not scale beyond a handful of workspaces and gives no delegated settings. **D** is a release mechanism.

Q9

SINGLE ANSWER

Notebook and semantic model changes must be reviewable in pull requests, with a full history of who changed what. What do you configure?

- A. A deployment pipeline with three stages.
- B. **Git integration** on the workspace, connected to an Azure DevOps or GitHub repository and branch.
- C. The XMLA endpoint set to Read Write.
- D. Impact analysis on each item.

ANSWER — B

Git integration is the version-control mechanism: commit from the workspace, update the workspace from the repo, and review diffs in the repository's normal pull-request flow.

A is the promotion mechanism and keeps no history of individual changes — this Git-versus-pipelines distinction is examined directly. **C** is a deployment surface, not version control. **D** analyses dependencies.

Q10**MULTI-SELECT — CHOOSE TWO****Which two statements about a Power BI Desktop project (.pbip) are correct?**

- A. The semantic model is saved as TMDL, a text format that can be diffed and merged.
- B. The report definition is saved as PBIR, also text-based.
- C. A .pbip contains a compressed copy of the imported data.
- D. A .pbip can only be opened by Tabular Editor.
- E. Saving as .pbip disables the XMLA endpoint for the model.

ANSWER — A AND B

A project is a folder of plain-text definitions — model as TMDL, report as PBIR — which is what makes genuine source control and code review of Power BI content possible.

C is what a .pbix does; a project separates definition from data. **D** and **E** are inventions. Do not confuse .pbip (a project) with .pbir (a template — everything except the data, for reuse) or .pbids (a data source file that opens Desktop pre-pointed at a source).

Q11**SINGLE ANSWER****You add a fourth workspace to an existing three-stage deployment pipeline for a pre-production stage. What is true about deployment pipelines?**

- A. Pipelines support exactly three stages.
- B. Pipelines support **2 to 10 stages**, and stages can be added, deleted and renamed.
- C. Pipelines support up to five stages, all of which must be named Dev, Test, UAT, Pre-prod, Prod.
- D. Additional stages require a separate pipeline per stage.

ANSWER — B

The default is three stages named Development, Test and Production, but the pipeline supports between 2 and 10 and they are freely nameable.

The related fact the exam probes is **pairing**: items in adjacent stages are paired when a workspace is assigned or when unpaired content is first deployed. Paired items overwrite their counterparts; unpaired items with the same name create duplicates instead of overwriting, and items added after assignment are not paired automatically.

Q12**SINGLE ANSWER****A semantic model must point at the Dev lakehouse in the Development stage and the Prod lakehouse in Production, and the same artefact must be promoted rather than rebuilt. What do you configure?**

- A. Three separate semantic models, one per stage.
- B. **Deployment rules** on the target stage, repointing the data source and parameters.
- C. A Git branch per stage.
- D. Separate capacities per stage.

ANSWER — B

Deployment rules are exactly this: per-stage overrides of data sources and parameter values, applied at deployment time so the promoted artefact is identical apart from where it points.

A is the thing pipelines exist to eliminate. **C** is version control, and a branch does not repoint a connection. **D** changes compute, not connections.

Q13**YES / NO SERIES**

Changes must be promoted from Development to Production without manual rebuild, and every change must also be auditable in source control. Proposed solution: connect the Development workspace to Git and use a deployment pipeline for promotion. Does this meet the goal?

ANSWER — YES

The two features are complementary and are designed to be used together: Git for version control and review, deployment pipelines for stage-to-stage promotion.

Variants in this series and their verdicts: "use Git integration alone and update the Production workspace from the main branch" (Yes, technically — Git can serve both roles, though it loses deployment rules and pairing); "use a deployment pipeline alone" (No — no change history); "export .pbix files and republish to Production" (No — manual, and not auditable).

Q14**SINGLE ANSWER**

Before dropping two columns from a warehouse table you must identify every downstream semantic model and report that would break, and notify their owners. What do you use?

- A. The workspace lineage view's **impact analysis** on the warehouse item.
- B. The Fabric Capacity Metrics app.
- C. The audit log.
- D. A deployment pipeline comparison.

ANSWER — A

Impact analysis is available on lakehouses, warehouses, dataflows and semantic models. It lists the dependent downstream items and reports, and lets you notify their contacts.

B reports capacity consumption. **C** records what has already happened. **D** compares stage contents, not dependencies. Whenever a stem says "before making a breaking change" or "identify affected reports", this is the answer.

Q15**MULTI-SELECT — CHOOSE TWO**

A DevOps pipeline must deploy a semantic model metadata change to Production without republishing the report. Which two are required?

- A. The capacity's **XMLA endpoint** must be set to Read Write.
- B. Deployment must be performed with an XMLA-capable tool such as Tabular Editor or ALM Toolkit.
- C. The workspace must be in My Workspace.
- D. The semantic model must be in DirectQuery mode.
- E. The report must be converted to a paginated report.

ANSWER — A AND B

XMLA read-write plus an XMLA client is the supported route for metadata-only deployment. It is also why Microsoft recommends enabling large semantic model storage format for models you deploy this way — XMLA write performance improves.

C is wrong in every direction; personal workspaces are excluded from most Fabric capabilities. **D** and **E** are unrelated to the deployment mechanism.

Q16

DRAG AND DROP — MATCH THE ASSET TO THE REQUIREMENT

Match each reusable asset to what it is for.

REQUIREMENT

ASSET

Give every team a starting report with the corporate theme, layout and measures, but no data

.pbit template

Let analysts open Power BI Desktop already connected to the approved warehouse, without typing a server name

.pbids data source file

Ensure forty reports all use one governed set of measures and relationships

Shared semantic model

ANSWER — AS TABULATED

Three different reuse mechanisms, and the exam expects you to distinguish them precisely: a template carries definition without data, a .pbids carries only a connection, and a shared semantic model carries the governed model itself.

The shared semantic model is the strongest of the three because it is a live dependency rather than a copy — which is why it pairs naturally with certification and with impact analysis.

Q17

SINGLE ANSWER

Self-service authors keep building near-duplicate models over the same warehouse, producing conflicting revenue figures. What is the best remedy?

- A. Remove workspace access from the self-service authors.
- B. Publish a **certified shared semantic model** and grant authors Build permission on it, so their reports connect live rather than importing their own.
- C. Apply a sensitivity label to the warehouse.
- D. Enable Git integration on the workspace.

ANSWER — B

One governed model, endorsed so it is discoverable and visibly trusted, with Build permission so people can create their own reports on top of it. That addresses both the duplication and the conflicting numbers.

A stops self-service rather than governing it — the exam favours enablement plus governance over prohibition. **C** and **D** address different problems entirely.

Q18**MULTI-SELECT — CHOOSE THREE****Which three actions can a workspace Contributor perform?**

- A. Create and edit lakehouses, notebooks and semantic models.
- B. Write data to items in the workspace.
- C. Read data in the workspace.
- D. Manage OneLake security roles.
- E. Delete the workspace.

ANSWER — A, B AND C

Contributor is the "build things" role: full content creation, read and write on data, no administration.

D is reserved for Admin and Member. **E** is Admin only. Learn the ladder as a table (see §3.9) — several questions in this domain are decided by one cell of it.

2 Prepare data

Q19 – Q48**Q19****SINGLE ANSWER**

A gold layer must be maintained by a team of T-SQL developers using stored procedures that UPDATE and DELETE rows, with multi-table transactional guarantees. Which item do you create?

- A. A Lakehouse and query it through the SQL analytics endpoint.
- B. A Warehouse.
- C. An Eventhouse.
- D. A Power BI datamart.

ANSWER — B

The Warehouse is the only Fabric analytics store with full T-SQL DML and DDL and multi-table transactions.

A fails on one decisive fact: a Lakehouse's SQL analytics endpoint is **read-only**. You write to a Lakehouse with Spark, not with T-SQL. If a stem mentions **INSERT**, **UPDATE**, **DELETE** or stored procedures that write, the answer is Warehouse. **C** is for event data. **D** is the legacy self-service option.

Q20**SINGLE ANSWER**

Ingestion must handle CSV, nested JSON and scanned PDF images, and the team writes PySpark. Which item is most appropriate for the bronze layer?

- A. A Warehouse.
- B. A Lakehouse.
- C. A KQL database.
- D. A Fabric SQL database.

ANSWER — B

A Lakehouse stores structured, semi-structured and unstructured data — it has a Files area as well as Tables — and its primary interface is Spark, which matches the team's skills.

A takes structured data only. C is optimised for telemetry and time-series. D is an operational database. The decision criteria the exam applies are data type, developer skill set, and whether T-SQL writes are needed.

Q21**SINGLE ANSWER**

Six million IoT readings per hour arrive from factory sensors. Analysts need sub-second queries over the last 30 days with time-bucketed aggregations. Which store?

- A. Warehouse
- B. Lakehouse
- C. Eventhouse / KQL database
- D. Semantic model in Import mode

ANSWER — C

An Eventhouse is purpose-built for high-volume, append-only, time-series and log data, queried with KQL and surfaced through Real-Time Dashboards.

The follow-up you should pre-empt: to make that data available to a Direct Lake semantic model, enable **OneLake integration** (also called OneLake availability) on the Eventhouse so its data is written to OneLake in Delta format — that is a named blueprint bullet.

Q22**SINGLE ANSWER**

Analytics is needed on an Azure SQL Database that backs a live application. The application team will not allow analytical queries against it, no ETL may be built, and the data must be at most minutes old. What do you implement?

- A. A nightly pipeline with a Copy activity.
- B. A DirectQuery semantic model against the Azure SQL Database.
- C. **Mirroring** of the Azure SQL Database into OneLake.
- D. A shortcut to the Azure SQL Database.

ANSWER — C

Mirroring continuously replicates a supported operational database into OneLake as Delta, near-real-time, with no pipeline to build and without putting analytical load on the source.

A fails the freshness and the "no ETL" requirements. **B** puts exactly the load on the source that the application team refuses. **D** is the trap: *external* shortcuts point at storage locations (ADLS, S3, GCS, Dataverse and similar), not at an external operational database such as Azure SQL DB — mirroring is the mechanism for those. An *internal* shortcut can target a Fabric SQL database, which is a different thing entirely.

Q23**SINGLE ANSWER**

A gold table in Workspace A must be queryable from a lakehouse in Workspace B. The data must not be duplicated and schema changes must appear automatically. What do you create?

- A. A pipeline that copies the table nightly.
- B. An **internal OneLake shortcut** in Workspace B's lakehouse pointing at the table.
- C. A Dataflow Gen2 with Workspace B's lakehouse as destination.
- D. A mirrored database.

ANSWER — B

A shortcut is a pointer: no copy, no ingestion latency, and schema changes in the source are reflected automatically.

"Without duplicating the data" is the most reliable keyword-to-answer mapping on this exam. **A** and **C** both create copies. **D** replicates an external operational database, which is not the scenario.

Q24**MULTI-SELECT — CHOOSE THREE****Which three are supported targets for an external OneLake shortcut?**

- A. Amazon S3 and S3-compatible storage
- B. Azure Data Lake Storage Gen2
- C. Google Cloud Storage
- D. An on-premises Oracle database via ODBC
- E. A Power BI report

ANSWER — A, B AND C

External shortcut targets are storage locations: ADLS Gen2, Azure Blob Storage, Amazon S3 and S3-compatible, Google Cloud Storage, Dataverse, Iceberg, OneDrive/SharePoint, and on-premises locations through the Fabric on-premises data gateway.

D is a relational database, not a storage location — that is an ingestion or mirroring scenario. **E** is not data storage at all. **Internal** shortcuts, by contrast, target Fabric items: lakehouses, warehouses, KQL databases, semantic models, SQL databases and mirrored catalogs.

Q25**YES / NO SERIES**

You must expose a folder of nested JSON files held in ADLS Gen2 inside a lakehouse so a notebook can read them. Proposed solution: create the shortcut under the lakehouse's Tables folder. Does this meet the goal?

ANSWER — NO

Shortcuts in the **Tables** folder can only be created at the top level and are intended for Delta-format tabular data, where schema and metadata sync automatically. Arbitrary file formats and nested hierarchies belong under **Files**, where shortcuts can be created at any level and no table discovery is attempted.

Variants: "create the shortcut under Files" (Yes); "copy the files in with a pipeline first" (Yes, but it duplicates data and loses the automatic sync); "create an external table in the Warehouse" (No — Fabric Warehouse does not take shortcuts this way). Also note the shortcut name restriction: no spaces for Tables-folder shortcuts, and no % or + characters anywhere.

Q26**SINGLE ANSWER**

A user with access to Lakehouse B, which contains a shortcut to a table in Lakehouse A, reports an error reading the shortcut. What is the most likely cause?

- A. The shortcut has expired.
- B. The user has no read permission on the **target** in Lakehouse A.
- C. The shortcut points at a Delta table.
- D. The lakehouses are in the same workspace.

ANSWER — B

Access through a shortcut requires permission in **both** the calling item and the target. Internal shortcuts authorise using the calling user's own identity, so an unpermitted user simply cannot read through it.

Shortcuts do not expire. The important exception the exam may test: when a semantic model uses Direct Lake on the SQL analytics endpoint, or T-SQL runs in delegated identity mode, the calling item owner's identity is passed instead of the end user's — which changes who is effectively authorised.

Q27

DRAG AND DROP — MATCH THE REQUIREMENT TO THE INGESTION ROUTE

Match each requirement to the most appropriate Fabric capability.

REQUIREMENT	CAPABILITY
Copy 60 tables nightly from on-premises SQL Server, then run a notebook only if the copy succeeds	Data pipeline
Business analysts must own and maintain a low-code transformation that lands in a lakehouse	Dataflow Gen2
Upsert 400 million rows into a Type 2 dimension using MERGE	Notebook (PySpark)
Query an existing Amazon S3 data set in place, with no copy	External shortcut
Keep an Azure SQL Database continuously replicated into OneLake with no pipeline	Mirroring

ANSWER — AS TABULATED

Pipelines orchestrate; dataflows and notebooks transform; shortcuts and mirroring get you access without building ingestion at all.

Choose between dataflow and notebook on **who owns it and how much data**: dataflows for analyst-owned, folding-friendly, moderate volumes; notebooks once the logic or the volume outgrows the mashup engine, or when the transformation must be unit-tested and version-controlled.

Q28

SINGLE ANSWER

An existing Azure Data Factory estate must be reused in Fabric with minimal rework. What is the most accurate statement?

- A. ADF pipelines cannot run in Fabric; everything must be rebuilt as notebooks.
- B. Fabric **Data Pipelines** use the same underlying engine and concepts as ADF, so activities, control flow and parameters transfer with limited rework; an existing ADF factory can also be mounted.
- C. ADF pipelines must be converted to Dataflow Gen2.
- D. ADF is only supported through a shortcut.

ANSWER — B

Data Pipelines are the Fabric surface of the Data Factory engine — Copy activity, control flow, parameters, triggers all behave familiarly, and mounting an existing ADF factory is supported as a deployment-pipeline item.

This matters practically as well as for the exam: prior ADF experience is directly transferable, and saying so is worth doing in an interview as well as recognising on a test.

Q29**SINGLE ANSWER****Which pair correctly describes the OneLake catalog and the Real-Time hub?**

- A. OneLake catalog discovers streaming sources; Real-Time hub discovers stored data.
- B. **OneLake catalog discovers and governs stored data items across the tenant; Real-Time hub discovers streaming sources and events.**
- C. Both are administrative dashboards for capacity consumption.
- D. Both are replaced by the lineage view.

ANSWER — B

Two discovery surfaces for two kinds of data, and the blueprint names both in the same bullet — "Discover data by using OneLake catalog and Real-Time hub".

A question here is usually recall, not reasoning. Learn the pair and move on.

Q30**SINGLE ANSWER****An Eventhouse holds three months of telemetry. A Direct Lake semantic model must read it. What must you do?**

- A. Export the KQL tables to CSV and import them into a lakehouse.
- B. **Enable OneLake integration (OneLake availability) on the Eventhouse so its data is written to OneLake in Delta format.**
- C. Create a DirectQuery model against the KQL database.
- D. Nothing — Direct Lake reads KQL databases natively.

ANSWER — B

OneLake integration makes Eventhouse data available in OneLake as Delta, which is the format Direct Lake requires. This is a named blueprint bullet: "Implement OneLake integration for Eventhouse and semantic models".

A duplicates data and loses freshness. **C** works but abandons the Direct Lake requirement in the stem. **D** is the misconception being tested.

Q31**SINGLE ANSWER****A shortcut must point at a folder on an on-premises file server. What is required?**

- A. Nothing — OneLake reaches on-premises locations directly.
- B. **The Fabric on-premises data gateway, and a cloud connection the shortcut creator has permission to use.**
- C. A personal-mode gateway on the creator's laptop.
- D. A mirrored database.

ANSWER — B

On-premises shortcut targets are reached through the Fabric on-premises data gateway, and external shortcuts delegate authorisation to a cloud connection — only users with permission on that connection may bind a shortcut to it.

C is Import-only and single-user; it is never the production answer. Note the separate fact that a **Direct Lake semantic model cannot use any gateway at all** — a constraint that appears in its own questions.

Q32**SINGLE ANSWER**

Gold-layer dimension tables must join efficiently to a 900-million-row fact and must support historical reporting. Which key design do you use?

- A. Composite natural keys of business code plus effective date, stored as strings.
- B. **Integer surrogate keys** generated at load, with the fact storing the surrogate that was current at transaction time.
- C. GUID keys generated by the source system.
- D. No keys; join on all descriptive columns.

ANSWER — B

Integer surrogates compress and join better than strings or GUIDs, and storing the surrogate that was current at transaction time is what makes Type 2 history resolve correctly.

A and **C** both create high-cardinality wide keys — the worst case for both Delta file layout and VertiPaq compression. GUIDs are additionally not supported as a Direct Lake semantic type.

Q33**SINGLE ANSWER**

A customer dimension must retain history when a customer's segment changes. Which implementation is correct?

- A. UPDATE the existing row with the new segment
- B. MERGE: close the current row (set ValidTo, IsCurrent = 0) and INSERT a new row with a new surrogate key, ValidFrom and IsCurrent = 1
- C. DELETE the row and INSERT a replacement
- D. Store the segment on the fact table instead

ANSWER — B

That is Type 2 slowly changing dimension handling. Implement it with **MERGE** — in T-SQL in a Warehouse, or against a Delta table from a notebook in a Lakehouse.

A is Type 1 and destroys history, so "sales by the customer's segment at the time of sale" becomes unanswerable. **C** is Type 1 with extra steps. **D** mixes dimension attributes into the fact and breaks the grain. The exam follow-up: "current segment" reporting on a Type 2 dimension needs a second path — filter `IsCurrent = TRUE` and re-map on the business key.

Q34

SINGLE ANSWER

A reusable transformation must be exposed to report authors as a queryable object, must accept a date-range parameter, and must live in a Fabric Warehouse. What do you create?

- A. A view.
- B. A table-valued function.
- C. A stored procedure.
- D. A materialised lake view in the Lakehouse.

ANSWER — B

A view cannot take parameters; a stored procedure can, but its result set cannot be joined to in a `SELECT`. A table-valued function is the object that is both parameterised and queryable.

The wider point the blueprint bullet is testing — "create views, functions and stored procedures" — is that all three are **Warehouse** capabilities. A Lakehouse SQL analytics endpoint supports views but is read-only, so no procedure there can write.

Q35

DROPDOWN COMPLETION

Complete the T-SQL that keeps only the most recent row per `CustomerKey` in a staging table.

```
WITH d AS (  
    SELECT *,  
           [ _____ ]() OVER (PARTITION BY CustomerKey  
                                ORDER BY UpdatedAt DESC, Id DESC) AS rn  
    FROM stg.Customer )  
SELECT * FROM d WHERE rn = 1;
```

- A. RANK
- B. DENSE_RANK
- C. ROW_NUMBER
- D. NTILE

ANSWER — C

`ROW_NUMBER` never ties, so exactly one row per key survives. `RANK` and `DENSE_RANK` both return 1 for tied rows, which would keep duplicates when `UpdatedAt` is identical.

Note the tie-breaker (, Id DESC) in the `ORDER BY`: without it the surviving row is non-deterministic. That detail turns up in well-written exam items and in real code reviews.

Q36

SINGLE ANSWER A silver table has nulls in `Segment` and duplicate rows on `CustomerKey`. Which PySpark snippet cleans both in a Lakehouse notebook?

- A. `df.dropna().distinct()`
- B. `df.dropDuplicates(["CustomerKey"]).fillna({"Segment": "Unknown"})`
- C. `df.filter(df.Segment.isNotNull())`
- D. `df.groupBy("CustomerKey").count()`

ANSWER — B

`dropDuplicates` with a subset deduplicates on the key rather than on the whole row, and `fillna` with a dictionary replaces nulls per column while keeping every row.

A `dropna()` deletes any row with a null anywhere — data loss the stem did not ask for — and `distinct()` deduplicates on all columns, which will not remove near-duplicate rows differing in one field. C deletes rows. D answers a different question.

Q37

SINGLE ANSWER A gold aggregate must be exposed to Power BI without users being able to change it, and must recompute automatically whenever the underlying fact is reloaded. In a Warehouse, what is the simplest object?

- A. A **view** over the fact and dimension tables.
- B. A physical table refreshed by a nightly pipeline.
- C. A Dataflow Gen2.
- D. A Power BI calculated table.

ANSWER — A

A view is always current by definition, requires no orchestration, and is read-only to consumers.

The trade-off the exam may push on: a view recomputes on every query, so at very large scale a materialised physical table refreshed by a pipeline can be the right answer instead — and a Direct Lake on *OneLake* model cannot read a SQL view at all, only Direct Lake on the SQL analytics endpoint can. Which option wins depends on the constraint in the stem.

Q38**SINGLE ANSWER**

You need every product in the catalogue that has never been sold. Which SQL construct is both correct and null-safe?

- A. `SELECT p.* FROM DimProduct p
WHERE p.ProductKey NOT IN (SELECT f.ProductKey FROM FactSales f);`
- B. `SELECT p.* FROM DimProduct p
WHERE NOT EXISTS (SELECT 1 FROM FactSales f
WHERE f.ProductKey = p.ProductKey);`
- C. `SELECT p.* FROM DimProduct p INNER JOIN FactSales f
ON f.ProductKey = p.ProductKey;`
- D. `SELECT p.* FROM DimProduct p FULL OUTER JOIN FactSales f
ON f.ProductKey = p.ProductKey;`

ANSWER — B

`NOT EXISTS` is null-safe and generally produces the better plan. A `LEFT JOIN ... WHERE f.ProductKey IS NULL` is the equivalent alternative.

A is the classic trap: if `FactSales.ProductKey` contains a single NULL, `NOT IN` returns no rows at all. C returns the opposite set. D returns everything.

Q39**SINGLE ANSWER**

A business analyst with no SQL must build a joined, filtered, grouped query over a warehouse and save it as a view. What do they use?

- A. The **Visual Query Editor** in the warehouse.
- B. Power Query in Power BI Desktop.
- C. A KQL queryset.
- D. A notebook with Spark SQL.

ANSWER — A

The Visual Query Editor is a no-code, Power Query-style canvas over a Lakehouse SQL analytics endpoint or a Warehouse. It generates SQL you can inspect, and the result can be saved as a view.

This is a named blueprint bullet — "Select, filter, and aggregate data by using the Visual Query Editor" — and "without writing code" in a stem points straight at it. B operates in the semantic-model layer, not in the warehouse.

Q40

DROPDOWN COMPLETION

Complete the KQL that returns average temperature in five-minute buckets per device over the last hour.

```
Telemetry
| [ __1__ ] Timestamp > ago(1h)
| [ __2__ ] avg(Temperature) by [ __3__ ](Timestamp, 5m), DeviceId
| order by Timestamp desc
```

- A. 1 = filter, 2 = aggregate, 3 = bucket
- B. 1 = where, 2 = summarize, 3 = bin
- C. 1 = select, 2 = group by, 3 = trunc
- D. 1 = where, 2 = project, 3 = bin

ANSWER — B

where filters, summarize ... by groups and aggregates, and bin(column, interval) buckets a timestamp.

D is the near-miss: project selects columns, it does not aggregate. The KQL vocabulary worth memorising for this exam is small — where, project, extend, summarize ... by, bin, ago, top N by, join kind=, take, render.

Q41

SINGLE ANSWER

A KQL database contains a shortcut named SalesShortcut. How do you query it?

- A. SalesShortcut | take 100
- B. external_table('SalesShortcut') | take 100
- C. SELECT TOP 100 * FROM SalesShortcut
- D. shortcut('SalesShortcut') | limit 100

ANSWER — B

Shortcuts in a KQL database appear in the Shortcuts folder and are treated as external tables, addressed with the external_table() function.

Contrast the other surfaces over the same shortcut: Spark reads it as

spark.read.format("delta").load("Tables/MyShortcut"), and the SQL analytics endpoint reads it as an ordinary table with SELECT TOP (100) * FROM [Lakehouse].[dbo].[MyShortcut]. Same data, three idioms.

Q42**SINGLE ANSWER** You must return revenue by product category for 2026 from a semantic model, as a query, without building a report. Which is correct?

- A. `SELECT Category, SUM(Revenue) FROM Model GROUP BY Category`
- B. `EVALUATE SUMMARIZECOLUMNS ('Product'[Category],
TREATAS ({ "2026" }, 'Date'[Year]),
"Revenue", [Total Revenue])`
- C. `Model | summarize sum(Revenue) by Category`
- D. `CALCULATE ( [Total Revenue], 'Date'[Year] = "2026" )`

ANSWER — B

A DAX query begins with `EVALUATE` and returns a table. `SUMMARIZECOLUMNS` is the grouping function Power BI itself generates, and `TREATAS` applies the year filter.

`D` is a measure expression, not a query — it returns a scalar and cannot be executed on its own. `A` and `C` are SQL and KQL against surfaces that do not exist here. Run DAX queries from the **DAX query view** in Desktop or through the XMLA endpoint.

Q43**SINGLE ANSWER** A T-SQL query must return, per product, the revenue of the previous month alongside the current month. Which construct do you use?

- A. `LAG(Revenue) OVER (PARTITION BY ProductKey ORDER BY MonthStart)`
- B. `ROW_NUMBER() OVER (ORDER BY MonthStart)`
- C. A self-join on `MonthStart = DATEADD(month, -1, MonthStart)`
- D. `CROSS APPLY` with a scalar subquery

ANSWER — A

`LAG` is the window function for "the previous row in this partition" and avoids the cost and fragility of a self-join.

`C` works but performs badly and silently drops products with a gap in their month series. Know the frame clause too: with duplicate ordering keys, `ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` and `RANGE ...` behave differently, which quietly changes a running total.

Q44**MULTI-SELECT — CHOOSE TWO**

A wide source table has 180 columns; reports use 22 of them. Which two actions best serve a Direct Lake gold table?

- A. Project only the required columns when writing the gold table.
- B. Convert the high-cardinality transaction datetime into a date column plus a time column.
- C. Store all 180 columns and hide the unused ones in the semantic model.
- D. Convert numeric columns to strings for consistency.
- E. Add a GUID surrogate key.

ANSWER — A AND B

Column count and cardinality drive both Delta file size and VertiPaq segment size. A datetime to the second has roughly 86,400× the cardinality of a date, so splitting it is usually the largest single win.

C hides columns from users but still stores and scans them. **D** makes compression worse. **E** is doubly wrong: GUIDs are high-cardinality, and the GUID semantic type is not supported in Direct Lake.

Q45**SINGLE ANSWER**

Source dates arrive as the text "2026-03-31T00:00:00Z" and must become a date column in the gold Warehouse table. Where should the conversion happen?

- A. In a DAX calculated column in the semantic model.
- B. In the **gold-layer transformation**, as an explicit `CAST / CONVERT` to `date`.
- C. In a report-level measure.
- D. Leave it as text and rely on implicit conversion.

ANSWER — B

Type conversion belongs upstream, where it is applied once, stored in the correct type, and available to every consumer of the table. It is a named blueprint bullet under Transform data.

A and **C** push work into the model and hide business logic from anything that is not Power BI — and a Direct Lake on SQL model cannot even have calculated columns. **D** defeats indexing, partition elimination and any date-based filter folding.

Q46**YES / NO SERIES**

A team must write cleaned rows back into a Lakehouse gold table on a schedule. Proposed solution: run a stored procedure on the Lakehouse SQL analytics endpoint from a pipeline. Does this meet the goal?

ANSWER — NO

The Lakehouse SQL analytics endpoint is **read-only**. It cannot execute DML, so no procedure there can write to the table. Writes to a Lakehouse are performed with Spark (a notebook or a Spark job definition) or by a Dataflow Gen2 with the lakehouse as its destination.

Variants: "run a notebook from the pipeline that writes with `saveAsTable`" (Yes); "use a Dataflow Gen2 with the lakehouse as destination" (Yes); "build the gold layer in a Warehouse and run the stored procedure there" (Yes — and it is often the better architecture when the team is SQL-native).

Q47**MULTI-SELECT — CHOOSE THREE**

Which three criteria does the Fabric decision guide use to choose between a Lakehouse and a Warehouse?

- A. The data type — structured only, or structured plus unstructured.
- B. The developer skill set — Spark or T-SQL.
- C. Whether T-SQL DML and multi-table transactions are required.
- D. Which storage format the data is held in.
- E. Whether the workspace is on an F or a P SKU.

ANSWER — A, B AND C

Those are the three axes that actually decide it.

D is the point of Fabric: *both* store Delta-Parquet in OneLake, so format is never the differentiator — recognising that is what the question is testing. **E** affects capacity limits and features, not the choice between these two items.

Q48**SINGLE ANSWER**

A fact table is loaded hourly by appending small files. After six weeks, Direct Lake queries have become noticeably slower although the row count has grown only modestly. What is the most likely cause and fix?

- A. The semantic model needs a full refresh.
- B. The Delta table has accumulated many small files, raising transcoding overhead. Run **OPTIMIZE** and enable auto compaction.
- C. The capacity needs to be upgraded.
- D. The DAX measures need variables.

ANSWER — B

Hourly appends produce many small files; each becomes more VertiPaq column segments, and transcoding cost rises. The diagnostic signal is file count growing faster than active table size, with files below the adaptive target of 128 MB – 1 GB.

A misunderstands Direct Lake — a "refresh" is metadata framing and copies no data, so it cannot fix a file-layout problem. **C** and **D** spend money and effort on the wrong layer. Remember the documented target: **1 to 16 million rows per row group**.

Q49**SINGLE ANSWER****What does a scheduled refresh of a Direct Lake semantic model actually do?**

- A. Copies all data from the Delta tables into the model's memory.
- B. Performs **framing** — analyses the latest Delta table metadata and updates the model's file references. No data is copied.
- C. Runs the Power Query steps against the source.
- D. Re-executes the DAX calculated columns.

ANSWER — B

Framing is metadata-only and takes seconds, which is why Direct Lake removes the refresh window and most of the refresh cost. Data reaches memory separately, at query time, through **transcoding** — only the columns and row groups the query needs.

The consequence worth stating: because loading is on demand, the volume of data a Direct Lake model can analyse may *exceed* the capacity's maximum memory. That is a genuine advantage over Import and a favourite exam fact.

Q50**SINGLE ANSWER****Row-level security is already defined at the warehouse's SQL analytics endpoint and must be enforced for all report users without being reimplemented in DAX. Which semantic model configuration do you use?**

- A. Direct Lake on OneLake.
- B. **Direct Lake on the SQL analytics endpoint.**
- C. Import with a scheduled refresh.
- D. Direct Lake on OneLake with a composite Import table.

ANSWER — B

Direct Lake on the SQL analytics endpoint goes through that endpoint and therefore **respects SQL-defined RLS** — falling back to DirectQuery where necessary to honour it.

A and **D** read the Delta files directly and consequently **bypass SQL-based RLS entirely** — which would be a security incident, not a design choice. **C** copies the data and detaches it from the source's security. This exact trade-off is the subject of its own blueprint bullet, so expect it more than once.

Q51**SINGLE ANSWER**

A Direct Lake model over the gold lakehouse must also contain a small Import table of manually maintained budget figures held in Excel. Which configuration supports this?

- A. Direct Lake on OneLake, in a composite model with the Import table.
- B. Direct Lake on the SQL analytics endpoint, in a composite model with the Import table.
- C. Either variant — both support composite models.
- D. Neither — Direct Lake models cannot be composite.

ANSWER — A

Composite models — mixing Direct Lake with Import tables — are supported only on the **OneLake** variant. Direct Lake on the SQL analytics endpoint cannot be combined with Import or DirectQuery tables.

Two further capability differences that follow the same split: Direct Lake on OneLake supports **calculated tables and calculated columns** (columns currently in preview) while the SQL variant supports neither; the SQL variant supports **SQL views** while the OneLake variant cannot connect to them. Calculation groups, field parameters and what-if parameters work on both.

Q52**SINGLE ANSWER**

In a development workspace you want any query that cannot be served by Direct Lake to fail loudly rather than quietly become slower. What do you set?

- A. Set `DirectLakeBehavior` to `DirectLakeOnly`, disabling DirectQuery fallback.
- B. Set `DirectLakeBehavior` to Automatic.
- C. Disable the SQL analytics endpoint.
- D. Reduce the capacity SKU so guardrails are hit sooner.

ANSWER — A

With fallback disabled the query returns an error instead of silently reverting to DirectQuery, which makes regressions visible. The documented recommendation is exactly this: fallback on in production for resilience, off in development to catch problems early.

B is the default and produces the silent slowdown the stem is trying to avoid. Fallback triggers to remember: a non-materialised SQL view, SQL-based granular access control, a guardrail breach, or an unprocessed table.

Q53**SINGLE ANSWER****A query scans more rows than the capacity's Direct Lake guardrail permits. What happens?**

- A. The same in both variants: the query fails.
- B. The same in both variants: the query falls back to DirectQuery.
- C. **It depends on the variant** — Direct Lake on SQL falls back to DirectQuery (if fallback is enabled); Direct Lake on OneLake has no fallback, so the refresh fails and the model cannot be queried until the Delta tables are optimised.
- D. The capacity is automatically scaled up.

ANSWER — C

This asymmetry is the practical consequence of the two architectures, and it is the reason the OneLake variant demands disciplined Delta table maintenance.

Guardrail shape worth remembering: F2-F32 all share a 300-million-row per-query ceiling, rising to 1,500 M at F64/P1 and doubling roughly per SKU tier thereafter. **Max model size** is evaluated at model level (10 GB at F2-F8, 20 GB at F16, 40 GB at F32, unlimited from F64/P1); everything else is per query.

Q54**MULTI-SELECT — CHOOSE THREE****Which three are true prerequisites or constraints for a Direct Lake semantic model?**

- A. It requires a Fabric capacity (F or P SKU); a Free licence is not sufficient.
- B. It cannot be created in a personal workspace (My Workspace).
- C. It cannot use an on-premises or VNET data gateway.
- D. It requires the semantic model and the data source to be in different regions.
- E. It requires incremental refresh to be configured.

ANSWER — A, B AND C

All three are documented constraints, and each has appeared as a distractor elsewhere.

D is the reverse of the truth — cross-region semantic model and data source is *not* supported. **E** is a category error: framing replaces refresh, so a Direct Lake table needs no incremental refresh policy at all.

Q55**SINGLE ANSWER**

A Direct Lake model returns errors on any visual that uses the Customer dimension. Investigation shows `DimCustomer.CustomerKey` contains duplicate values. Why does this fail rather than merely give wrong totals?

- A. Because Direct Lake does not support relationships.
- B. Because Direct Lake requires the **one side** of a relationship to contain unique values; duplicates cause queries to fail.
- C. Because the key is an integer.
- D. Because RLS is not configured.

ANSWER — B

Direct Lake enforces two relationship rules strictly: related columns must have matching data types, and the one-side column must contain unique values. A duplicate on the one side produces a query failure, not a silent miscount.

The fix is upstream, in the gold layer: deduplicate the dimension before it lands. Related unsupported types to remember for Direct Lake: binary and GUID semantic types, complex Delta column types, non-numeric floating point values, and strings longer than 32,764 characters.

Q56**SINGLE ANSWER**

A gold Delta table is written by a nightly Spark job and read only by a Direct Lake model and by report queries. Which write-time optimisation should be enabled?

- A. **V-Order**, at table level via `delta.parquet.vorder.enabled`.
- B. VACUUM with a one-hour retention.
- C. Nothing — V-Order is on by default in every workspace.
- D. Partition the table by a high-cardinality key.

ANSWER — A

V-Order reorganises Parquet row-group distribution, encoding and compression at write time. It costs roughly 15% more on write and pays back substantially on read — precisely the trade you want for a read-heavy analytics table.

C is the currency trap: V-Order is now **disabled by default in new Fabric workspaces**, so it must be enabled deliberately at session, table or write level. **B** is dangerous — shortening VACUUM retention can remove versions that time travel and concurrent readers still need. **D** over-partitioning is a classic small-file generator.

Q57**MULTI-SELECT — CHOOSE TWO**

A table receives frequent updates and deletes but few small-file appends. Queries have degraded. Which two maintenance actions are appropriate?

- A. Run **OPTIMIZE** periodically to purge deletion vectors from qualifying files.
- B. Schedule **VACUUM** separately from **OPTIMIZE** to reclaim storage from unreferenced files.
- C. Set **VACUUM** retention to zero to reclaim space immediately.
- D. Disable auto compaction.
- E. Convert the table to CSV.

ANSWER — A AND B

OPTIMIZE purges a file's deletion vectors once more than **5%** of its records are marked deleted; auto compaction only does so when the partition also meets its small-file trigger, which is why an update/delete-heavy table still needs periodic **OPTIMIZE**. **VACUUM** runs on its own schedule because it addresses storage, not layout.

C risks breaking time travel and concurrent readers. **D** removes the recommended default maintenance for Spark-written tables. **E** abandons Delta and therefore Direct Lake.

Q58**SINGLE ANSWER**

An Import semantic model is expected to grow to 14 GB in the service and will be deployed from Azure DevOps using Tabular Editor. What must you enable?

- A. **Large semantic model storage format** on the model (or as the workspace default).
- B. Incremental refresh.
- C. Direct Lake fallback.
- D. Automatic page refresh.

ANSWER — A

Without the large format a model is capped at 1 GB; with it, the model can grow to the capacity limit, which is required to exceed 10 GB. Microsoft additionally recommends enabling it for any model deployed through the XMLA endpoint because XMLA write performance improves.

B shortens each refresh but does not raise the size ceiling — though it is a sensible companion. Note the related fact: the Power BI Desktop *upload* limit remains 10 GB; the model grows past it in the service after refresh.

Q59**MULTI-SELECT — CHOOSE THREE****Which three are required to configure incremental refresh on an Import fact table?**

- A. Two parameters named exactly `RangeStart` and `RangeEnd` , of type Date/Time.
- B. A filter on the fact query's date column between those parameters.
- C. The filter must fold to the source.
- D. The table must be in Direct Lake storage mode.
- E. A calculated column flagging recent rows.

ANSWER — A, B AND C

The names are fixed and case-sensitive, the filter must use them, and — the part people forget — **the filter must fold**, or the service reads everything on every partition and the policy makes matters worse.

D is a contradiction: framing replaces refresh in Direct Lake. **E** would break folding. Optional extras worth knowing: **detect data changes** to skip unchanged partitions, and a **hybrid table** with a DirectQuery partition on the leading edge for near-real-time.

Q60**SINGLE ANSWER****A 4-billion-row DirectQuery fact must support fast summary reporting with drill-through to detail. What do you build?**

- A. Import the whole fact table.
- B. A **composite model**: a DirectQuery detail fact plus an Import aggregation table, with shared dimensions set to **Dual** storage mode.
- C. Two separate semantic models.
- D. DirectQuery with automatic page refresh.

ANSWER — B

Aggregations let summary queries hit an in-memory table while only drill-through reaches the source. Dual dimensions are essential — they let the engine serve each dimension from memory or from the source depending on which fact the query touches.

A exceeds what is reasonable to import. **C** splits the user experience. **D** adds refresh frequency, not speed. Note the constraint for the Fabric variant of this question: composite models require **Direct Lake on OneLake**, not the SQL endpoint variant.

Q61**SINGLE ANSWER**

Thirty base measures each need Current, PY, YoY % and YTD variants, and the YoY % variant must display as a percentage while the others display as currency. What do you build?

- A. 120 individual measures with format strings.
- B. A **calculation group** with four calculation items using `SELECTEDMEASURE()`, and a **dynamic format string** on the YoY % item.
- C. A field parameter listing the 30 measures.
- D. A what-if parameter.

ANSWER — B

```
-- Calculation item: "YoY %"  
VAR C = SELECTEDMEASURE()  
VAR P = CALCULATE ( SELECTEDMEASURE(), SAMEPERIODLASTYEAR ( 'Date'[Date] ) )  
RETURN DIVIDE ( C - P, P )  
-- with a dynamic format string on the item: "0.0%"
```

C is the distractor the blueprint deliberately sets up by naming calculation groups, dynamic format strings and field parameters in one bullet: a **field parameter** lets a user *swap* which measure or dimension a visual shows — a different problem. **D** generates a disconnected numeric table for scenario analysis.

Q62**SINGLE ANSWER**

Users must be able to switch a chart between Revenue, Margin and Units, and between Category, Region and Channel, from two slicers, without you building nine visuals. What do you implement?

- A. A calculation group per axis.
- B. Two **field parameters** — one over the three measures, one over the three dimensions.
- C. Bookmarks with selected visuals.
- D. Personalize visuals.

ANSWER — B

Field parameters generate a table of fields that a slicer can drive, swapping what the visual displays. Two parameters give the full 3×3 matrix from one chart.

A modifies calculations, not field selection. **C** would need nine states and nine visuals. **D** lets consumers personalise for themselves but does not give the author a controlled, curated list — and the stem implies a designed experience.

Q63

DROPDOWN COMPLETION

Complete the DAX that returns the previous month's revenue on each row of a matrix, using a window function rather than time intelligence.

```
Prev Month Revenue =
CALCULATE (
    [Revenue],
    [ _____ ] ( -1, , ORDERBY ( 'Date'[MonthStart], ASC ) )
)
```

- A. INDEX
- B. OFFSET
- C. WINDOW
- D. ROWNUMBER

ANSWER — B

OFFSET(delta, ...) returns the row a given number of positions before or after the current one. A delta of -1 is the previous row in the ordering.

A INDEX returns an absolute position (first, last, nth). **C** WINDOW returns a range of rows — use it for running totals and moving averages. **D** ROWNUMBER returns a position number. All five share the parameters relation, ORDERBY, PARTITIONBY, MATCHBY and blanks, and you must supply either relation or ORDERBY.

Q64

SINGLE ANSWER

A rank measure shows "1" on the grand-total row of a matrix, which is meaningless. Which information function fixes it?

- A. ISBLANK
- B. IF (HASONEVALUE ('Product'[Name]), [Rank])
- C. ISERROR
- D. CONTAINSSTRING

ANSWER — B

HASONEVALUE is TRUE only when exactly one value of the column is in the current filter context — false on a total row — so the measure returns blank there and the total disappears.

The blueprint names "information functions" explicitly. The set worth knowing: HASONEVALUE, SELECTEDVALUE (the shorthand for IF(HASONEVALUE(c), VALUES(c), alt)), ISINSCOPE (which level of a hierarchy is being evaluated), ISBLANK, ISFILTERED / ISCROSSFILTERED, and USERPRINCIPALNAME.

Q65

SINGLE ANSWER

A measure is slow. You must determine whether the time is spent in the storage engine or the formula engine, without leaving Power BI Desktop. What do you use?

- A. Performance Analyzer to isolate the visual, then the **DAX query view** to run and tune the copied query.
- B. The Fabric Capacity Metrics app.
- C. Impact analysis.
- D. The lineage view.

ANSWER — A

Performance Analyzer splits each visual into DAX query, visual display and other; "Copy query" hands you the DAX; the DAX query view lets you run it, use **DEFINE MEASURE** to test a rewrite, and commit only when the timing improves. The blueprint names both tools together.

DEFINE

```
MEASURE Sales[Revenue v2] = SUMX ( Sales, Sales[Qty] * Sales[Price] )
```

EVALUATE

```
SUMMARIZECOLUMNS ( 'Date'[Year], "Rev", [Revenue v2] )
```

For the FE/SE split itself, DAX Studio's Server Timings is the classic external tool — but the exam names Performance Analyzer and the DAX query view, so answer with those unless an external tool is explicitly offered.

Q66

MULTI-SELECT — CHOOSE TWO

A measure over a 500-million-row fact is slow. Which two changes are most likely to help?

- A. Replace `CALCULATE([Sales], FILTER(Sales, Sales[Qty] > 10))` with `CALCULATE([Sales], Sales[Qty] > 10)`.
- B. Remove a measure reference from inside a **SUMX** over the fact, to avoid per-row context transition.
- C. Turn the measure into a calculated column.
- D. Set all relationships to bidirectional.
- E. Wrap the expression in **IFERROR**.

ANSWER — A AND B

A converts a materialised table scan into a storage-engine predicate. **B** removes context transition executed once per row — one of the most common causes of formula-engine-bound measures.

C stores data permanently and enlarges every scan, and is impossible on Direct Lake on SQL anyway. **D** creates ambiguity and more work. **E** hides errors and blocks optimisation.

Q67**SINGLE ANSWER**

A shared semantic model has 40 dependent reports. You must change a measure's definition and deploy it to Production without republishing any report. What do you do?

- A. Edit the model in Desktop and republish the .pbix, overwriting the model and reports.
- B. Run **impact analysis**, then deploy the metadata change through the **XMLA endpoint** with Tabular Editor or ALM Toolkit.
- C. Create a new semantic model and repoint each report manually.
- D. Use a deployment pipeline to copy the reports.

ANSWER — B

Impact analysis tells you which reports are affected and lets you notify their owners; the XMLA endpoint deploys a metadata-only change to the model without touching the reports that connect to it.

A republishes everything and risks overwriting report changes made in the service. **C** is 40 manual edits. **D** promotes items but is not the mechanism for a metadata-only model change. This question deliberately spans two domains — the exam does that.

05 Case study

One scenario, five questions — the format that punishes people who read the questions before the requirements.

How to work a case study. Read the requirements tabs first; almost every answer is decided by one sentence in them. You may review questions **within** the case, but once you move past it you cannot return. Budget 12–15 minutes for five questions and do not over-think the easy ones — every case contains one or two.

Existing environment

Northwind Logistics India runs 240 depots. Shipment data lives in an on-premises SQL Server (**ShipDB**): a **FactShipment** table of 620 million rows over nine years, plus Depot, Customer, Route and Status dimensions. Sensor telemetry from depot conveyors arrives at roughly 5 million events per hour. Finance maintains an annual target workbook in Excel on SharePoint at month × depot grain. A partner shares reference data as Parquet files in an **Amazon S3** bucket that Northwind is contractually forbidden to copy. The tenant has one **F64** capacity. Today all 400 analysts hold workspace **Member** access to a single workspace. The data engineering team writes PySpark; the finance systems team writes T-SQL and maintains stored procedures.

Requirements

- R1.** Depot managers must see only their own depot's rows. Row-level security is already defined at the SQL analytics endpoint and must not be reimplemented in DAX.
- R2.** The partner's S3 reference data must be queryable in Fabric without being copied.
- R3.** Conveyor telemetry must be queryable with sub-second latency for the last 30 days, and the same data must also be readable by a Direct Lake semantic model.
- R4.** The finance systems team must own the gold layer using stored procedures that update rows.
- R5.** Before any breaking change to a gold table, the affected downstream reports must be identified and their owners notified.
- R6.** Administrative effort must be minimised and analysts must not be able to modify published content.

Q68

SINGLE ANSWER Which semantic model configuration satisfies R1?

- A. Import mode with DAX row-level security roles.
- B. Direct Lake on OneLake.
- C. **Direct Lake on the SQL analytics endpoint.**
- D. DirectQuery against **ShipDB**.

ANSWER — C

Only the SQL analytics endpoint variant respects row-level security defined in SQL. R1 says explicitly that the security must not be reimplemented in DAX, which rules out both Import with DAX roles and any option requiring new role definitions.

B reads the Delta files directly and therefore **bypasses** SQL-defined RLS — a security failure, not a design trade-off. **D** would honour source security but abandons the performance and platform benefits, and puts analytical load on an on-premises OLTP database. Accept the consequence of C: no composite models and no calculated columns in that model.

Q69**SINGLE ANSWER****Which approach satisfies R2 for the partner's S3 data?**

- A. A pipeline with a Copy activity from S3 into a lakehouse.
- B. An **external OneLake shortcut** to the S3 bucket, created in the lakehouse Files or Tables area.
- C. Mirroring the S3 bucket.
- D. A Dataflow Gen2 reading from S3 into a warehouse.

ANSWER — B

Amazon S3 and S3-compatible storage are supported external shortcut targets. A shortcut creates no copy, adds no ingestion latency and synchronises schema changes automatically — which is exactly what a "must not be copied" contractual constraint requires.

A and **D** both copy the data and breach the contract. **C** mirrors operational databases, not object storage. Practical detail: the shortcut binds to a cloud connection, and only users with permission on that connection can create shortcuts using it.

Q70**MULTI-SELECT — CHOOSE TWO****Which two actions satisfy R3 for the conveyor telemetry?**

- A. Ingest the telemetry into an **Eventhouse / KQL database** via an Eventstream.
- B. Enable **OneLake integration** on the Eventhouse so its data is written to OneLake in Delta format.
- C. Import the telemetry into a Power BI semantic model every 15 minutes.
- D. Store the telemetry as CSV in the lakehouse Files area.
- E. Create a shortcut from the lakehouse to **ShipDB**.

ANSWER — A AND B

The Eventhouse gives sub-second KQL queries over high-volume append-only event data; OneLake integration then makes the same data available as Delta so a Direct Lake model can read it without a second copy. That combination is precisely the blueprint bullet "Implement OneLake integration for Eventhouse and semantic models".

C caps out at 48 refreshes a day and cannot deliver sub-second freshness. **D** abandons both the query performance and the Delta requirement. **E** addresses a different source entirely.

Q71**SINGLE ANSWER****Where should the gold layer live to satisfy R4?**

- A. A Lakehouse, with the finance team writing stored procedures on its SQL analytics endpoint.
- B. A **Warehouse**, with the finance team writing T-SQL stored procedures, views and functions.
- C. An Eventhouse.
- D. A Power BI datamart.

ANSWER — B

R4 requires stored procedures that *update rows*, and the finance team is T-SQL-native. Only the Warehouse offers full T-SQL DML and DDL with multi-table transactions.

A is the distractor the case is built around: a Lakehouse SQL analytics endpoint is **read-only**, so no procedure there can update anything. Bronze and silver can still sit in a Lakehouse — the PySpark team owns those — with gold in the Warehouse. Both store Delta in OneLake, so the split costs nothing downstream.

Q72**MULTI-SELECT — CHOOSE TWO****Which two actions address R5 and R6 together?**

- A. Run **impact analysis** on the gold warehouse before schema changes, and notify the listed downstream owners.
- B. Move the 400 analysts from workspace **Member** to consuming published content, and reduce any remaining workspace access to **Viewer**.
- C. Give all 400 analysts the Contributor role so they can fix their own reports.
- D. Apply a Confidential sensitivity label to every item.
- E. Create 240 separate workspaces, one per depot.

ANSWER — A AND B

B is the step candidates most often miss, and it also retro-fixes Q68: while the analysts are workspace Members they have full data access and **bypass RLS entirely**, so the SQL-defined security in R1 would do nothing for them. Reducing them to Viewer or app consumption is what makes the rest of the design real.

C moves in the wrong direction on both requirements. D classifies and protects but grants and removes nothing, so it cannot satisfy either R5 or R6. E is administrative effort maximised.

06

Final-week checklist, method and sources

What to verify before you book, and where every fact in this guide comes from.

6.1 Can you do all of these without looking anything up?

Maintain a data analytics solution

- ☐ Recite the four workspace roles and which can manage OneLake security
- ☐ Say who OneLake security roles actually constrain, and why
- ☐ Distinguish Read / ReadData / ReadAll / Write item permissions
- ☐ Map a requirement to RLS, CLS, OLS or file-level security
- ☐ Say what a sensitivity label does and does not do
- ☐ Distinguish Promoted, Certified and Master data, and who may apply each
- ☐ Explain domains and delegated administration
- ☐ Configure Git integration on a workspace and say how it differs from a pipeline
- ☐ Explain .pbip, TMDL and PBIR
- ☐ State the deployment pipeline stage range and what pairing means
- ☐ Use deployment rules to repoint a source per stage
- ☐ Run impact analysis and say when it is the answer
- ☐ Deploy a model through the XMLA endpoint and name the prerequisite setting
- ☐ Distinguish .pbit, .pbids and a shared semantic model

Prepare data

- ☐ Choose between Lakehouse, Warehouse, Eventhouse and SQL database on the three real criteria
- ☐ State that a Lakehouse SQL analytics endpoint is read-only
- ☐ Choose between pipeline, Dataflow Gen2, notebook, shortcut and mirroring
- ☐ List the external shortcut targets and where shortcuts may be created
- ☐ Explain shortcut permission behaviour in both items
- ☐ Say when mirroring beats a pipeline
- ☐ Distinguish the OneLake catalog from the Real-Time hub
- ☐ Enable OneLake integration on an Eventhouse and say why
- ☐ Build a gold star schema with integer surrogate keys
- ☐ Implement SCD Type 2 with MERGE in T-SQL and in PySpark
- ☐ Deduplicate with ROW_NUMBER and handle nulls with fillna
- ☐ Write the same aggregation in SQL, in KQL and as a DAX query
- ☐ Query a shortcut from KQL with external_table()
- ☐ Say what the Visual Query Editor is for

Semantic models

- ☐ Explain framing and transcoding in one sentence each
- ☐ Choose between Direct Lake on OneLake and on the SQL analytics endpoint, and name four differences
- ☐ Explain DirectLakeBehavior and when to disable fallback
- ☐ Say what happens on a guardrail breach in each variant
- ☐ List three Direct Lake prerequisites or constraints
- ☐ Explain why duplicate one-side keys cause failures
- ☐ Say what V-Order, OPTIMIZE, Z-Order and VACUUM each do
- ☐ Diagnose a small-file problem and quote the row-group target
- ☐ Say when large semantic model storage format is required
- ☐ Configure incremental refresh and state the folding requirement
- ☐ Build a composite model with Dual dimensions
- ☐ Distinguish calculation groups, dynamic format strings and field parameters
- ☐ Use OFFSET, WINDOW, INDEX, RANK and ROWNUMBER
- ☐ Use HASONEVALUE, SELECTEDVALUE and ISINSCOPE
- ☐ Diagnose a slow measure with Performance Analyzer and the DAX query view

6.2 The week before

1. **Take the official practice assessment cold** and work only on what scores below 75%.
2. **Spend a full day in a trial capacity** building both Direct Lake variants over the same gold layer. Nothing on this exam repays hands-on time like that comparison does.
3. **Write ten lines of KQL from memory** — `where` , `summarize by bin` , `project` , `extend` , `top by` , `external_table` . Two hours, several marks.
4. **Read §3.9 aloud**. The security domain is pure recall and is the cheapest place to lose marks by vagueness.
5. **Re-check the published skills list** the day before. The blueprint is versioned — the current one is effective 21 July 2026 — and it does move.
6. **Try the exam sandbox** so the interactive question widgets are familiar, and test your equipment the day before if you are sitting online-proctored.

6.3 Method and caveats

- The syllabus in §2 reproduces the Microsoft Learn DP-600 study guide as published, showing the skills measured as of **21 July 2026**. Weightings are Microsoft's; the commentary after each bullet is this guide's interpretation.
- Question counts are not published by Microsoft. The "≈ questions of 45" figures in §0.2 apply the published percentage bands to a hypothetical 45-question form and are an illustration only.
- The 72 questions in §4 and §5 are **original**, written against the published blueprint and documented product behaviour. They model the exam's style; they are not real exam items and no real item is reproduced here.
- Direct Lake guardrail figures, capability differences and limitations are quoted from the Microsoft Learn Direct Lake overview as current in September 2026. Fabric ships continuously — features described as preview may be generally

available by the time you sit the exam, and guardrail numbers can change. Verify anything that decides an answer.

- V-Order's default state has changed over time; the figure used here — **disabled by default in new workspaces** — is the current documented behaviour and is exactly the kind of detail worth re-checking before exam day.
- Pricing is indicative and regional; the figure shown at booking is authoritative.

6.4 Sources

Microsoft Learn — [Study guide for Exam DP-600](#) (skills measured as of 21 July 2026, audience profile, change log) · Microsoft Learn — [Microsoft Certified: Fabric Analytics Engineer Associate](#) credential page (duration, languages, renewal, retake, SQL/KQL/DAX prerequisite) · Microsoft Learn — [Direct Lake overview](#) (framing and transcoding, the OneLake and SQL analytics endpoint variants, `DirectLakeBehavior` and fallback, capacity guardrail table, relationship and data-type limitations, infrastructure constraints) · Microsoft Learn — [Get started with OneLake security](#) (workspace role matrix, item permissions, OneLake security roles, RLS/CLS) · Microsoft Learn — [Fabric decision guide: choose between Lakehouse and Warehouse](#) · Microsoft Learn — [Delta Lake table optimization and V-Order](#) (V-Order behaviour and default state) · Microsoft Learn — [Cross-workload table maintenance and optimization](#) (OPTIMIZE, auto compaction, deletion vectors, VACUUM cautions, row-group targets, small-file diagnosis) · Microsoft Learn — [OneLake shortcuts](#) (internal and external targets, Tables vs Files placement, permissions, limits) · Microsoft Learn — [Introduction to deployment pipelines](#) (stage range, supported items, pairing, relationship to Git) · Microsoft Learn — [Large semantic model storage format](#) · Microsoft Learn — [OFFSET function \(DAX\)](#) and the window-function family · Microsoft Learn — [Exam scoring and score reports](#), [Exam retake policy](#), [Certification renewal](#).

The one habit that decides this exam. DP-600 questions almost always have two options that work and one constraint that eliminates one of them. Train yourself to find the constraint before you read the options — *without duplicating the data, must support T-SQL updates, security defined in SQL must be enforced, least privilege, minimise administrative effort*. On this paper the constraint is the question; the technology is just the vocabulary it is written in.